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SPEED DETERMINATION IN ROBOTICS SYSTEMS AND APPLICATIONS

Abstract

In various examples, a technique for generating speed change decisions for a mobile robot includes identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of the mobile robot in the physical environment. The technique also includes generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path. The technique further includes generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

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Background/Summary

BACKGROUND

[0001] To traverse an environment, many autonomous or semi-autonomous mobile robots or other machine types move through the environment to a given destination according to a route or path determined using a path planner. An efficient route should minimize the time and/or distance of travel to the destination, should safely avoid static and moving obstacles, and should be generated in a short amount of time to avoid delays between receiving mission instructions and executing the mission.

[0002] Existing approaches for route or path planning use a route or path planner to find a route or path plan that specifies an efficient route from a starting point to a destination point. The route plan includes a sequence of waypoints, each of which corresponds to a location in the physical environment of a mobile robot. The robot moves to each waypoint in the sequence to reach the destination point. In existing approaches, route plans can cover distances that are relatively short such as 5-10 meters, and longer distances such as 1 kilometer or more. The existing approaches do not scale well to such longer distances, however. For example, in existing approaches, the route planner searches for a route from the start to the destination. For a longer distance between the start and the destination, such as 1 kilometer or more, processing the numerous possible routes can be an intractable searching problem that is impractical to solve in a short amount of time-which may limit these types of approaches for real-time or near real-time deployments.

[0003] Further, existing approaches do not account for changes in speed that are made by the robot as it encounters regions of different types that affect the speed at which the robot moves while executing the mission. Further, existing route planning approaches do not take interactions with dynamic obstacles into account when determining route plans. The interactions with dynamic obstacles, such as automatic doors that can open and close, can cause the robot to adjust its speed while executing the mission. Since existing route planning approaches do not take such regions and dynamic obstacle interactions into account, portions of the route that pass through regions of different types and/or regions having dynamic obstacles can take more time to traverse than expected by the route planner. The route planner can incorrectly select route plans having such regions instead of other route plans that would have lower travel time and/or shorter distance when executed by the robot, e.g., if the other route plans have longer travel time or distance in comparison to route plans determined without taking into account speed reductions the robot makes when it encounters regions of different types and/or dynamic obstacles while executing a mission. Still further, in existing approaches, the path planner has limited extensibility and adaptability to changes in the robot system. For example, existing path planners do not readily accommodate different drivetrain configurations or different space representations. Accordingly, existing path planners are difficult to adapt to different kinematics and maneuvering capabilities.

[0004] As such, a need exists for more effective techniques for generating route plans for long distances and/or for environments having regions of different types that affect the speed at which a robot moves in autonomous or semi-autonomous systems.

SUMMARY

[0005] Embodiments of the present disclosure relate to generating speed change decisions for a mobile robot. The techniques described herein include identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of the mobile robot in the physical environment. The techniques also include generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path. The techniques further include generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed

profile of the mobile robot to satisfy the one or more speed constraints.

[0006] One technical advantage of the disclosed techniques relative to prior approaches is the ability to efficiently navigate large environments because, for example, fine-grain path planning is performed on individual regions on a route plan instead of the substantially larger area between the start and destination locations of the route plan. Further, Frenet coordinate discretization may be used in the regions to improve the efficiency of path searching by focusing the path search on a space near the path. Accordingly, the disclosed techniques are faster and less resource-intensive than prior approaches that use a single path planner to generate the route. Another technical advantage of the disclosed techniques is the ability to dynamically adjust speed using semantic information from map labels and real-time or near real-time perception, such as predicted positions of dynamic obstacles. Adjusting the speed in this way results in smoother robot motion, for example. Accordingly, the disclosed techniques allow the robot to navigate more safely and efficiently through the environment than prior approaches that do not use semantic information to dynamically adjust speed. These technical advantages represent one or more technological improvements over prior approaches.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present systems and methods for route or path planning for robotics systems and applications are described in detail below with reference to the attached drawing figures, wherein:

[0008] FIG. 1 illustrates a computing device configured to implement one or more aspects of various embodiments;

[0009] FIG. 2 illustrates a robot navigation system that includes the regional path planner and speed decision maker of FIG. 1, according to various embodiments;

[0010] FIG. 3 is a more detailed illustration of the regional path planner of FIG. 1, according to various embodiments;

[0011] FIG. 4A illustrates a navigable space that includes a mobile robot, waypoints, obstacles, and a regional path between waypoints, according to various embodiments;

[0012] FIG. 4B illustrates a navigable space in which a target waypoint is invalid and candidate waypoints are identified as potential replacements for the invalid waypoint, according to various embodiments;

[0013] FIG. 5 illustrates a flow diagram of a method for generating a regional path, according to various embodiments;

[0014] FIG. 6 is a more detailed illustration of the speed decision maker of FIG. 1, according to various embodiments;

[0015] FIG. 7 illustrates a navigable space that includes static and dynamic obstacles, and a path-time space representation of the navigable space, according to various embodiments;

[0016] FIG. 8 illustrates a flow diagram of a method for generating speed directives for a path, according to various embodiments;

[0017] FIG. 9A is an illustration of an example autonomous vehicle, in accordance with some embodiments of the present disclosure;

[0018] FIG. 9B is an example of camera locations and fields of view for the example autonomous vehicle of FIG. 9A, in accordance with some embodiments of the present disclosure;

[0019] FIG. 9C is a block diagram of an example system architecture for the example autonomous vehicle of FIG. 9A, in accordance with some embodiments of the present disclosure;

[0020] FIG. 9D is a system diagram for communication between cloud-based server(s) and the example autonomous vehicle of FIG. 9A, in accordance with some embodiments of the present disclosure;

[0021] FIG. **10** is a block diagram of an example computing device suitable for use in implementing some embodiments of the present disclosure; and

[0022] FIG. **11** is a block diagram of an example data center suitable for use in implementing some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0023] Systems and methods are disclosed for regional path planning in robotic systems and applications. Although the present disclosure can be described with respect to an example autonomous or semi-autonomous vehicle or machine **900** (alternatively referred to herein as “vehicle **900**” or “ego-machine **900**,” an example of which is described with respect to FIGS. **9A-5D**), this is not intended to be limiting. For example, the systems and methods described herein can be used by, without limitation, non-autonomous vehicles or machines, semi-autonomous vehicles or machines (e.g., in one or more adaptive driver assistance systems (ADAS)), autonomous vehicles or machines, piloted and un-piloted robots or robotic platforms, warehouse vehicles, off-road vehicles, vehicles coupled to one or more trailers, flying vessels, boats, shuttles, emergency response vehicles, motorcycles, electric or motorized bicycles, aircraft, construction vehicles, underwater craft, drones, and/or other vehicle types. In addition, although the present disclosure can be described with respect to robotic systems and applications, this is not intended to be limiting, and the systems and methods described herein can be used in augmented reality, virtual reality, mixed reality, robotics, security and surveillance, autonomous or semi-autonomous machine applications, and/or any other technology spaces where sensor monitoring can be used.

[0024] FIG. **1** illustrates a computing device **100** configured to implement one or more aspects of various embodiments. In at least one embodiment, computing device **100** includes a desktop computer, a laptop computer, a smart phone, a personal digital assistant (PDA), a tablet computer, a server, one or more virtual machines, an embedded system, a system on a chip, a computing system of an autonomous, semi-autonomous, or a non-autonomous machine, and/or any other type of computing device configured to receive input, process data, and optionally display information, and is suitable for practicing one or more embodiments. Computing device **100** is configured to run a regional path planner **122** and a speed decision maker **124** that can reside in a memory **116**. It is noted that the computing device described herein is illustrative and that any other technically feasible configurations fall within the scope of the present disclosure. For example, multiple instances of regional path planner **122** and/or speed decision maker **124** can execute on a set of nodes in a distributed and/or cloud computing system to implement the functionality of computing device **100**. Alternatively, computing device **100** can be implemented similar to that of the computing device of the example autonomous or semi-autonomous machine **500** described at least with respect to FIGS. **9A-9D**.

[0025] In at least one embodiment, computing device **100** includes, without limitation, an interconnect (bus) **112** that connects one or more processors **102**, an input/output (I/O) device interface **104** coupled to one or more input/output (I/O) devices **108**, memory **116**, a storage **114**, and/or a network interface **106**. Processor(s) **102** can include any suitable processor implemented as a central processing unit (CPU), a graphics processing unit (GPU), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA), an artificial intelligence (AI) accelerator, a deep learning accelerator (DLA), a parallel processing unit (PPU), a data processing unit (DPU), a vector or vision processing unit (VPU), a programmable vision accelerator (PVA), any other type of processing unit, or a combination of different processing units, such as a CPU(s) configured to operate in conjunction with a GPU(s). In general, processor(s) **102** can include any technically feasible hardware unit capable of processing data and/or executing software applications. Further, in the context of this disclosure, the computing elements shown in computing device **100** can correspond to a physical computing system (e.g., a system in a data center or a machine) and/or can correspond to a virtual computing instance executing within a computing cloud.

[0026] In at least one embodiment, I/O devices **108** include devices capable of receiving input, such as a keyboard, a mouse, a touchpad, a VR/MR/AR headset, a gesture recognition system, a steering wheel, mechanical, digital, or touch sensitive buttons or input components, and/or a microphone, as well as devices capable of providing output, such as a display device, haptic device, and/or speaker. Additionally, I/O devices **108** can include devices capable of both receiving input and providing output, such as a touchscreen, a universal serial bus (USB) port, and so forth. I/O devices **108** can be configured to receive various types of input from an end-user (e.g., a designer) of computing device **100**, and to also provide various types of output to the end-user of computing device **100**, such as displayed digital images or digital videos or text. In some embodiments, one or more of I/O devices **108** are configured to couple computing device **100** to a network **110**.

[0027] In at least one embodiment, network **110** is any technically feasible type of communications network that allows data to be exchanged between computing device **100** and internal, local, remote, or external entities or devices, such as a web server or another networked computing device. For example, network **110** can include a wide area network (WAN), a local area network (LAN), a wireless (e.g., WiFi) network, a cellular network, and/or the Internet, among others.

[0028] In at least one embodiment, storage **114** includes non-volatile storage for applications and data, and may include fixed or removable disk drives, flash memory devices, and CD-ROM, DVD-ROM, Blu-Ray, HD-DVD, or other magnetic, optical, or solid-state storage devices. Regional path planner **122** and/or speed decision maker **124** may be stored in storage **114** and loaded into memory **116** when executed.

[0029] In one embodiment, memory **116** includes a random-access memory (RAM) module, a flash memory unit, and/or any other type of memory unit or combination thereof. Processor(s) **102**, I/O device interface **104**, and network interface **106** may be configured to read data from and write data to memory **116**. Memory **116** may include various software programs or more generally software code that may be executed by processor(s) **102** and application data associated with said software programs, including regional path planner **122** and/or speed decision maker **124**.

[0030] Regional path planner **122** includes functionality to generate a regional path that specifies a path for the mobile robot to follow to navigate from a start waypoint toward a target waypoint while avoiding obstacles. The regional path does not necessarily end at the target waypoint. The regional path can end at a location different from the target waypoint if, for example, the target waypoint is occupied by an obstacle, or the regional path planner **122** is unable to identify a path to the target waypoint. The start and target waypoints can be locations on a global route plan that covers a longer distance than the regional path. As such, the regional path planner **122** performs path planning for individual regions that are each smaller than the area between the start and destination locations of the global route plan. The terms “route” and “path” as used herein may both refer to a sequence of locations, e.g., waypoints, that can be used by a mobile robot to navigate from a starting location, such as the current location of the mobile robot, to a target location. The term “route” is used herein for longer distances than the term “path” for explanatory purposes. A route can be represented by the same information as a path in some implementations, or by different information than a path in other implementations. Further, the terms “global,” “regional,” and “local” are used for explanatory purposes and do not necessarily imply specific distances or area sizes. In some examples described herein, a global route plan can cover multiple kilometers, a regional path can cover approximately 50 meters, and a local trajectory can cover approximately 5 meters, but the global route plan, regional path, and local trajectory can cover any suitable distances in other examples.

[0031] The regional path planner **122** is configured to generate a regional path for a mobile robot from input that includes a global route plan represented as a sequence of waypoints, a current location of the mobile robot, a distance map, and/or a semantic map of the physical environment of the mobile robot. The distance map can indicate the distances to obstacles in the physical environment, and can be provided by a perception system that uses sensors to identify the locations

of the obstacles. The semantic map, which can be used as an alternative to or in addition to the distance map, indicates locations of static (e.g., non-moving) obstacles in the physical environment. The regional path planner **122** identifies a search space that includes at least a portion of the global route plan, e.g., a portion of the global route plan between the current location of the mobile robot and a region planning horizon. The regional planning horizon can correspond to a distance for which regional path planning is to be performed. The regional path planner **122** identifies a target waypoint in the sequence of waypoints. The target waypoint can be a farthest waypoint on the global route plan that is not beyond the planning horizon.

[0032] The regional path planner **122** uses the distance map and/or semantic map to determine whether the target waypoint is occupied by an obstacle. If the target waypoint is occupied, the regional path planner **122** identifies a replacement target waypoint. The replacement target waypoint can be identified by searching along the direction of the path or laterally to the path, e.g., in a direction perpendicular to the path at the current location of the mobile robot. The regional path planner **122** uses a path search algorithm to generate a path from the current location to the target waypoint, or to the replacement target waypoint if the target waypoint is occupied by an obstacle. If the path search algorithm is unable to find a collision-free path, the regional path planner **122** expands the search space and searches again, or requests that the global route plan be modified to use a different route, and attempts to find a collision-free path using the modified global route plan. The mobile robot can then navigate to the target waypoint via the path, and the regional path planner **122** can be used repeatedly to find next target waypoints and regional paths to the next target waypoints, until the mobile robot navigates to the end (e.g., destination waypoint) of the global route plan.

[0033] Speed decision maker **124** includes functionality to generate a set of speed directives that specify how the speed of the mobile robot is to be changed to avoid collisions with obstacles on the regional path. The speed decision maker **124** generates speed constraints using input that includes a regional path, a distance map, and/or a semantic map. The distance map and/or the semantic map indicate the locations of obstacles in the physical environment and also include semantic information about obstacles and/or about regions of the environment. The information about obstacles can include a type of each obstacle and a specification of motion of the obstacle. The information about regions can include a speed limit associated with a region, for example.

[0034] The speed decision maker **124** uses the distance map and/or semantic map to identify one or more obstacles on or near a given path in the physical environment. An obstacle is near the given path if the obstacle is within a threshold distance of the path, for example. The speed decision maker **124** generates one or more speed constraints based on characteristics of the identified obstacles. Each speed constraint is associated with one or more map regions that include at least one of the identified obstacles. The speed decision maker **124** generates one or more speed change decisions that change the speed of the robot over time, such that the determined speeds of the robot over time minimize a total cost determined by the constraints. A constraint can determine a cost if the speed of the robot exceeds a speed limit associated with the constraint when the robot is within a speed zone on or near the path during one or more specified time intervals. The speed change decisions can specify actions such as accelerating, decelerating, or stopping. Since the path is given as an input, the locations along the path at which speed constraints are to be enforced can be specified as one-dimensional distances, e.g., as start and end distances. For moving obstacles, the times during which speed constraints are to be enforced can be specified as ranges of elapsed travel time, e.g., as start and end times.

[0035] The speed decision maker **124** can cause each speed change decision to be applied to the mobile robot at a respective time. For obstacles that move over time, the constraints can specify location ranges (e.g., a portion of the path) and time ranges (e.g., a time interval) to which speed limits apply. The speed limit can be 0 to indicate that the mobile robot is to stop at particular location(s) during a particular time range. Since there can be multiple moving obstacles, multiple

speed constraints can apply during the same time period and at the same location. Accordingly, a solver is used to generate speed change decisions that satisfy the speed constraints, e.g., by minimizing a total cost determined by the constraints. The solver can search a path-time space for speed profiles that satisfy the constraints, for example. The solver can alternatively or additionally divide the path-time space into discrete intervals, and search for a solution in each interval, for example. The speed change decisions generated by the speed decision maker **124** can be converted to speed directives suitable for input to a trajectory planner. The speed directives can specify speed limits that apply during particular regions and times, and the trajectory planner can generate trajectories that satisfy the speed directives.

[0036] FIG. **2** illustrates a robot navigation system **200** that includes the regional path planner **122** and speed decision maker **124** of FIG. **1**, according to various embodiments. The robot navigation system **200**. The robot navigation system **200** includes a regional path planner system **202** and system components, such as a mapping system **204**, a global route planner **208**, a localization system **212**, and a perception system **220**, which are related to robot navigation and provide input to the regional path planner system **202**. One or more of the system components **204**, **208**, **212**, **220** can be located external to a mobile robot, e.g., on a cloud-based server that communicates with the mobile robot via a communications network. The regional path planner system **202** is located on the mobile robot in examples described herein, although one or more components of the regional path planner system **202** can be external to the mobile robot in other examples. The regional path planner system **202** generates a local trajectory **232** for the mobile robot to use in navigation along a portion of a global route plan received from the global route planner **208**.

[0037] The mapping system **204** provides a semantic map **206** specifying locations of obstacles and/or navigable regions in the physical environment of the mobile robot. The semantic map **206** also specifies semantic information associated with the obstacles and/or navigable regions, such as obstacle types. The semantic map **206** represents the physical environment of the mobile robot and includes semantic information about objects in the physical environment such as obstacles, which are to be avoided by the mobile robot so that collisions do not occur. The semantic map **206** includes information about static obstacles that do not move. The semantic map **206** represents the physical environment as a set of semantic regions identified by geographic coordinates or other location information.

[0038] The semantic information can include a region type that characterizes a region. A region type can be associated with each region to indicate whether the region is a navigable region, through which the robot can navigate without collision, or an obstacle to be avoided by the robot. A region type can also indicate further details about an obstacle, such as an obstacle type. Possible obstacle types include, e.g., wall, stairs, or other static (e.g., non-movable) obstacles. Other possible obstacle types include automatic door, pedestrian, vehicle, or other dynamic (e.g., movable) obstacle. For dynamic obstacles, the semantic information can further specify how the obstacle moves over time, e.g. as a range of times during which the object occupies its associated region (e.g., a closed automatic door), and/or a range of times during which the obstacle does not occupy its associated region and the robot can occupy the associated region (e.g., an open automatic door). The semantic map **206** can be represented, for example, as a graph in which nodes represent semantic regions and edges represent connections between semantic regions.

[0039] The global route planner **208** generates a global route plan between a starting point, such as the current location of the mobile robot, and a destination point. The global route plan is represented as a sequence of global route waypoints **210**, which include a starting global route waypoint and a destination global route waypoint.

[0040] The localization system **212** generates localization information of the mobile robot using a localization map, which can be the semantic map **206** or other suitable map of the physical environment. The localization information includes the position (e.g., (x, y, z) coordinates) and/or orientation (e.g., rotation about the x axis, rotation about the y axis, and rotation about the z axis) of

the mobile robot. The localization information is represented as a robot pose **214**. The robot pose **214** specifies the location and/or orientation of the mobile robot in the physical environment. The localization system **212** uses sensors located on the mobile robot to identify distances to and locations of objects in the physical environment, and uses the distances and locations to find the location of the mobile robot on the localization map. The sensors can include, e.g., camera(s), RADAR, LiDAR, and/or the like.

[0041] The perception system **220** generates a distance map **224** using sensors located on the mobile robot, e.g., camera(s), RADAR, LiDAR, and/or the like. The distance map **224** provides locations of obstacles within detection range of the sensors. The distance map **224** can be a grid-based representation of the physical environment in which the mobile robot is located. The obstacle locations provided by the distance map **224** can include locations of obstacles that are not necessarily present in the semantic map **206**, such as dynamic obstacles that move over time and are identified in real time, e.g., during navigation of the mobile robot.

[0042] The regional path planner system **202** includes a semantic region extractor **222**, the regional path planner **122** and the speed decision maker **124** of FIG. 1, a local trajectory planner **230**, and a robot controller **234**. The semantic region extractor **222** identifies one or more navigable regions **216** that intersect the global route and/or are within a threshold distance of the global route using input that includes the navigable regions **216**, the robot pose **214**, and the global route waypoints **210**. At least one of the navigable regions **216** includes the mobile robot so that the mobile robot can reach the path by navigating through the navigable regions **216**. For example, the global route planner **208** can have a planning horizon of more than 100 meters up to multiple kilometers. A global route can be generated once per robot mission. The semantic region extractor **222** can have a planning horizon of approximately 100 meters, e.g., greater than a sensor detection range, and can operate at a frequency such as 1 cycle per second, where each cycle corresponds to an invocation of the semantic region extractor **222**. The regional path planner **122** can have a planning horizon of approximately 50 meters, e.g., the sensor detection range, and can operate at a frequency such as 1 cycle per second, where each cycle corresponds to an invocation of the regional path planner **122**. The local trajectory planner **230** can have a planning horizon of approximately 5 meters and can operate at a frequency such as 10 cycles per second, where each cycle corresponds to an invocation of the local trajectory planner **230**.

[0043] The regional path planner **122** generates a regional path **226** using input that includes the navigable regions **216** identified by the semantic region extractor **222**, the robot pose **214**, and the distance map **224**. The regional path **226** can include a sequence of waypoints of a path to be followed to navigate from a current location of the mobile robot to a next waypoint of the global route waypoints **210**, if such a path can be found. If the regional path planner **122** does not find such a path, the regional path planner **122** can identify a different destination waypoint or request that the global route planner **208** generate a different global route plan and provide corresponding different global route waypoints **210**. The regional path planner **122** uses a path solver algorithm, e.g., the A* algorithm or the like, to find the regional path **226**.

[0044] The speed decision maker **124** generates one or more speed directives **228** using input that includes the distance map **224**, the navigable regions **216**, and the regional path. The navigable regions **216** can be map(s), e.g., portions of the semantic map **206**, that represent regions of the physical environment. The navigable regions **216** include regions of the physical environment in which the mobile robot is permitted to navigate without constraints. The navigable regions **216** can also include speed zones that are associated with speed limits. Speed zones can be regions that include stairs, doorways, ramps, annotated high-risk areas, or the like, for example. The speed limit associated with a zone can be zero to indicate that navigation in the speed zone is prohibited. The distance map **224** identifies obstacles, and the speed decision maker **124** identifies which of the obstacles are on or near (e.g., within a threshold distance of) the regional path **226**. The distance map **224** can include semantic information associated with the obstacles, such as obstacle types.

Alternatively or additionally, the speed decision maker **124** can also use the semantic map **206** to identify speed zones and/or obstacles on or near the regional path **226** and semantic information associated with the obstacles.

[0045] The speed directives **228** specify speed changes at particular times and/or distances, such as acceleration, deceleration, stopping, and so on. The speed directives **228** represent a “speed profile,” in which is a specification of speed over time (e.g., a function that represents values of speed over a time range). Since the obstacles on the regional path **226**, such as pedestrians, automatic doors, and so on, can move while the mobile robot is traversing the regional path **226**, there can be a number of possible local optimums that correspond to different potential choices for the speed profile. The speed decision maker **124** can evaluate these possible local optimums and identify one that best satisfies criteria such as minimizing travel time, minimizing travel distance, or other criteria.

[0046] The local trajectory planner **230** generates a local trajectory **232** using input that includes the regional path **226** and the speed directives **228**. The local trajectory planner **230** uses optimization techniques to generate the local trajectory **232** by optimizing the regional path **226** and/or the speed directives **228**. If both the regional path **226** and the speed directives **228** are provided as input to the local trajectory planner **230**, the local trajectory planner **230** optimizes the regional path **226** and the speed directives **228** jointly to generate the local trajectory **232**. The local trajectory **232** can differ from the regional path **226** and speed directives **228** because the optimizations may change the waypoints of the regional path **226** and/or the speed directives **228**. The local trajectory **232** can be a sequence of points (e.g., waypoints), for example. The robot controller **234** causes the robot to navigate in the physical environment according to the local trajectory **232**, e.g., by actuating wheels and/or other components of a drive train.

[0047] FIG. **3** is a more detailed illustration of the regional path planner **122** of FIG. **1**, according to various embodiments. The regional path planner **122** generates a regional path **348** from a current location of the mobile robot to a destination based on a next waypoint in a sequence of global route waypoints **210**. The regional path planner **122** includes a path problem formulator **302**, a path solver **314**, and a failure handler **350**. The path problem formulator **302** generates solver input to be provided to the path solver **314**. The solver input includes one or more candidate next states **312** for the mobile robot. The term “state” is used herein to refer to a location, or a move or motion that causes the mobile robot to travel to the location. Accordingly, the candidate next state(s) **312** identify potential next locations for the mobile robot. The solver input also includes a next state cost **332** for each candidate next state **312**, and a valid target waypoint **340**, which represents a candidate destination for the regional path **348**. The path solver **314** attempts to generate a regional path **348** using the solver input.

[0048] If the path solver **314** successfully generates a regional path **348**, then the regional path **348** is provided to the local trajectory planner **230**. If the path solver **314** is unable to generate a regional path **348**, e.g., because the possible paths are occupied by obstacles, then a failure handler **350** attempts to generate a fallback regional path **352**. If the failure handler **350** is unable to generate a fallback regional path **352**, e.g., by expanding the size of the search space, then the failure handler **350** can request that the global route plan be re-generated (e.g., re-routed), and the regional path planner **122** can use the path solver **314** to attempt to find a fallback regional path **352** using a different sequence of global route waypoints **210**.

[0049] The path problem formulator **302** uses the robot model **304**, space discretizer **316**, cost manager **328**, and waypoint manager **334** to generate the solver input. The robot model **304** provides information about properties of the robot, including a robot shape **306**, available motion operations, and one or more robot constraints **308**. The robot constraints **308** include maximum speed, maximum acceleration, and so on. The robot model **304** also includes a candidate next state generator **310**, which generates candidate next state(s) **312** based on the robot constraints **308** and a discretized space **318**. The space discretizer **316** generates the discretized space **318** by dividing the

navigable regions **216** into smaller regions such as cells. The size of each cell is determined by a discretization resolution. The discretized space **318** is associated with a coordinate system in which locations in the discretized space **318** can be identified by coordinate values. The discretization resolution can be used to convert between coordinate values and distances in the physical environment. The coordinate system can be a Cartesian coordinate system or a Frenet coordinate system, for example. If a Cartesian coordinate system is used, the discretized space **318** can be implemented as a two-dimensional grid **320A** that divides the space into square (or rectangular) cells. The grid **320A** has an x axis and a y axis. Each cell is identified by an (x, y) coordinate pair that specifies a location of the cell relative to the origin of the coordinate system. The space discretizer **316** can generate the grid **320A** by dividing the navigable regions **216** into a number of rectangular cells, which form the grid **320A**.

[0050] If the Frenet coordinate system is used, the discretized space **318** can be implemented as a two-dimensional lattice **320B** that divides the space into cells. A location in Frenet coordinates has an (s, d) coordinate pair, where s specifies a longitudinal displacement along a reference path and d specifies a lateral displacement from the reference path. In other words, the s coordinate specifies a distance along the reference path (such as the global route plan that passes through the global route waypoints **210**). The d coordinate specifies a side-to-side position relative to the reference path on an axis perpendicular to the reference path at the distance specified by the s coordinate of an (s, d) pair. Each cell can be identified by an (s, d) coordinate pair that specifies the location of the cell relative to a starting point on the reference path (e.g., relative to s=0).

[0051] The space discretizer **316** also generate a discretized space graph **322**, which includes a set of nodes **324** that correspond to locations in the discretized space **318**. For example, each node **324** can correspond to one or more cells of the cell grid **320** of the discretized space **318**. The discretized space graph **322** also includes a set of edges **326**. Each edge **326** connects a pair of nodes **324** that correspond to adjacent cells of the discretized space **318**. Adjacent cells represent locations that are adjacent in the physical environment. That is, an edge **326** that connects two nodes **324** in the discretized space graph **322** indicates that the mobile robot can navigate between the locations represented by the two nodes **324** (at least if the destination location is not occupied).

[0052] The candidate next state generator **310** uses the discretized space graph **322** to identify the candidate next states **312** for the mobile robot. The candidate next state(s) **312** can include robot motions or instructions that specify how the mobile robot is to move to navigate to a candidate location. The candidate location can be represented as a node **324** of the discretized space graph **322**, and corresponds to a cell of the discretized space **318**. Alternatively or additionally, the candidate next state(s) **312** can include the candidate locations. For example, the candidate next state(s) **312** can include nodes **324** that represent the candidate locations.

[0053] The candidate next state generator **310** generates the candidate next state(s) **312** by finding nodes **324** in the discretized space graph **322** that represent possible next states for the mobile robot. To generate each candidate next state **312**, the candidate next state generator **310** searches the discretized space graph **322** to identify nodes **324** representing locations that are between a current location of the mobile robot and a next global route waypoint of the global route waypoints **210**. The next global route waypoint is a next one of the global route waypoints **210** that is past the current location of the mobile robot in the direction of the destination (e.g., end) global route waypoint. The candidate next state generator **310** verifies that the transition to each of the identified nodes **324** is valid, e.g., that the mobile robot can move to the location represented by the identified node **324** without violating the robot constraints **308**, and that the identified node **324** is not occupied by an obstacle. For each identified node **324** that is valid, the candidate next state generator **310** generates a candidate next state **312**. The candidate next state(s) **312** can include robot motions or instructions that specify how the mobile robot is to move to navigate to a candidate location. Alternatively or additionally, the candidate next state(s) **312** can include the candidate locations.

[0054] The cost manager **328** computes a next state cost **332** for each candidate next state(s) **312**. The next state cost **332** is computed based on navigation objectives, which include robot kinematics objectives, global route guidance objectives, and/or obstacle avoidance objectives. The next state cost **332** is computed as a sum of robot kinematics costs, global route guidance costs, obstacle avoidance costs, and/or other costs that can be computed, e.g., by extensions to the cost manager.

[0055] The robot kinematics costs penalize infeasible motions. The goal of the robot kinematics costs is to minimize the discrepancy between the regional path **348** and the local trajectory **232** generated by the local trajectory planner **230**. The penalized motions can include lateral acceleration and jerk (e.g., rate of change of acceleration over time). Infeasible lateral acceleration and jerk can be penalized by using the maximum speed and acceleration provided by the robot model **304** to approximate the speed profile.

[0056] The global route guidance cost encourages the robot to stay on the global route plan specified by the global route waypoints **210**. The global route guidance cost can be based on the distance to the global route plan, which can be calculated by generating a distance map for the global route plan represented by the global route waypoints **210**. The global route plan could be infeasible with respect to real-time obstacles. Accordingly, the penalty imposed by the global route guidance cost should be relatively small, such that the mobile robot can properly avoid obstacles by deviating from the global route plan. However, the mobile robot should not move too far away from the global route plan, as it might lose sight of the global guidance (e.g., landmark obstacles). As such, a limit can be imposed on the amount of deviation from the global route plan.

[0057] The costs computed for obstacle avoidance objectives penalize obstacle collisions. For example, obstacle avoidance objectives can incur a high cost for grid areas occupied by obstacles. The regional path planner **122** obtains obstacle information from the navigable regions **216** and/or from the distance map **224** provided by the perception system **220**, and computes costs for the obstacles based on distances to the obstacles. The obstacle locations are provided by the navigable regions **216** and/or the distance map **224**, e.g., as obstacle polygons. The distance to an obstacle can be computed as the distance between the polygon and the node **324** in which the mobile robot is currently located.

[0058] The waypoint manager **334** generates a valid target waypoint **340**, which is provided to the path solver **314** for use as a destination of the regional path **348**. To generate the valid target waypoint **340**, the waypoint manager **334** searches the global route waypoints **210** for a candidate target waypoint **336**, which can be a waypoint in the global route waypoints **210** that is between the current location of the mobile robot and the final destination waypoint of the global route waypoints **210**. The waypoint manager **334** uses a waypoint validator **338** to determine whether the candidate target waypoint **336**. The waypoint validator **338** determines that the candidate target waypoint **336** is valid if the location represented by the candidate target waypoint **336** is not occupied by an obstacle or other obstruction, and is within a planning horizon. The planning horizon can correspond to the sensor detection range of the mobile robot, for example.

[0059] If the waypoint validator **338** determines the candidate target waypoint **336** is valid, then the waypoint manager **334** outputs the target waypoint **336** as a valid target waypoint **340**. If the waypoint validator **338** determines that the candidate target waypoint **336** is invalid, then the waypoint manager **334** searches for a replacement target waypoint, e.g., using a longitudinal search and/or a lateral search for a valid “point” along the global route plan or lateral to the global route plan. The term “point” as used herein can refer to a location or region in the discretized space **318**, e.g., a cell on a grid **320A** or lattice **320B**. A point can correspond to an amount of space represented by a cell of the discretized space **318**, for example. The global route waypoints **210** are examples of points. The components of the regional path planner **122**, such as the waypoint manager **334** and/or the path solver **314**, can select points that are not included in the global route waypoints **210** and use the selected points in at least a portion of the regional path. For example, if

one of the global route waypoints **210** is occupied by an obstacle, the waypoint manager **334** finds a different point and uses the different point as a target waypoint **340** for the regional path **348**. Accordingly, the replacement target waypoint can be one of the global route waypoints **210**, or another valid point that is not one of the global route waypoints **210**. As an example, the replacement target waypoint can be the first valid point found by searching in a longitudinal and/or lateral direction from the invalid candidate target waypoint **336**. The replacement target waypoint can be the first such valid point found by the search. If a valid point is not found, e.g., there are no valid points along the global route plan starting from the invalid candidate target waypoint **336** in the longitudinal direction, and no valid points in a lateral direction, then waypoint manager **334** reports a failure.

[0060] To perform a longitudinal search, the waypoint manager **334** starts from the invalid candidate target waypoint **336** and searches the points along the global route plan until the first valid point is found. The first valid point found, if any, is used as the replacement target waypoint **336**. The longitudinal search approach is likely to progress toward the destination, but can also cause repetitive searching when a valid point is located beyond the planning horizon. To perform a lateral search, the waypoint manager **334** starts from the invalid candidate target waypoint **336** and searches along a direction perpendicular to the route plan until the first valid point is found. The first valid point found, if any, is used as the replacement target waypoint **336**. The lateral search approach can help ensure that the identified valid point is within the planning horizon, but could introduce larger deviations from the original global route plan than the longitudinal search approach. The waypoint manager **334** outputs the replacement target waypoint, if found, as the valid target waypoint **340**. The waypoint manager **334** reports a failure if a valid target waypoint **340** is not found.

[0061] The path problem formulator **302** provides the candidate next state(s) **312**, the next state cost **332**, and the valid target waypoint **340** to the path solver **314**. The path solver **314** uses a path searcher **344** to search for a regional path **348** from a current location of the mobile robot to the valid target waypoint **340** via the candidate next state(s) **312**. The path searcher **344** can use a path searching algorithm such as the A* algorithm, which uses a heuristic based on Euclidean distance to the valid target waypoint **340**. Alternatively or additionally, the path searcher **344** can use a bidirectional A* algorithm, a weighted A* algorithm, a D* Lite algorithm, or other suitable path searching algorithm.

[0062] If the path searcher **344** successfully finds a regional path **348**, then the regional path **348** is provided to the local trajectory planner **230**. If the path searcher **344** is unable to find a regional path **348**, e.g., because the possible paths are occupied by obstacles, then the path solver **314** provides a solver status **346** indicating a failure to the failure handler **350**. Upon receiving the solver status **346** indicating failure, the failure handler **350** attempts to cause the path solver **314** to generate a fallback regional path **352**. The failure handler **350** can expand the search space and use the path solver **314** to search the expanded search space for a fallback regional path **352**.

Alternatively, the failure handler **350** can use the global route planner **208** to generate a new global route plan having one or more new global route waypoint(s) **210**. The path problem formulator **302** generates adjusted solver input based on the new global route waypoint(s) **210**. The adjusted solver input includes new candidate next state(s) **312**, a new next state cost **332**, and a new valid target waypoint **340**. The path solver **314** then attempts again to find a fallback regional path **352** using the adjusted solver input. If the fallback regional path **352** is successfully generated, the failure handler **350** provides the fallback regional path **352** to the local trajectory planner **230**. The local trajectory planner **230** uses the regional path **348** or fallback regional path **352** to generate a local trajectory **232** for the mobile robot to follow. The local trajectory **232** can be a sequence of points in 2D or 3D space, for example.

[0063] FIG. 4A illustrates a navigable space **400** that includes a mobile robot **402**, global route waypoints **404**, obstacles **414**, **414**, and a regional path **410** between global route waypoints **404**,

according to various embodiments. A global route plan from a starting point to a destination point is represented by a starting global route waypoint **404A**, intermediate global route waypoints **404B**, **404C**, and a destination global route waypoint **404D**. Each global route waypoint **404A**, **404B**, **404C**, **404D** is located in a respective navigable region **406A**, **406B**, **406C**, **406D**. A mobile robot **402** is located in the navigable region **406B**. The mobile robot **402** has a sensing range **416** shown as a circle centered on the mobile robot **402**. Mapped obstacles **412A**, **412B** are located outside the navigable regions **406**. The locations and boundaries of the mapped obstacles **412** are identified from a semantic map **206**. An unmapped obstacle **414** is located in navigable regions **4046B** and **406C**. The location and boundaries of the unmapped obstacle **414** are provided by a distance map **224**. A regional path **410** has been generated for the mobile robot **402**. A local trajectory **408** has also been generated, e.g., based on the regional path **410** or on a previous regional path (not shown). The regional path **410** is from the location of the mobile robot **402** to the global route waypoint **404C**, which is near the limit of the sensing range **416**. The regional path **410** is from the location of the mobile robot **402** to a destination waypoint, which can be a global route waypoint **404**, that is within or near the sensing range **416** so that unmapped obstacles **414** detected by sensors are avoided by the regional path planner **122** in generating the regional path **410**.

[0064] FIG. 4B illustrates a navigable space **400** in which a target waypoint **436** is invalid and candidate waypoints **440** are identified as potential replacements for the invalid waypoint, according to various embodiments. A global route plan is represented by sequence of waypoints including a starting global route waypoint **432**, global route waypoints **404A**, **404B**, and a destination global route waypoint **442** received from a global route planner **208**. A mobile robot **402** has navigated along a portion of the global route plan and is located between the starting global route waypoint **432** and the global route waypoint **404A**. The mobile robot **402** has a planning horizon **430**, beyond which planning is not performed because of a limited sensing range. The regional path planner **122** attempts to find a regional path from the location of the mobile robot **402** toward the destination global route waypoint **442**. The regional path planner **122** can evaluate global route waypoints **404** received from a global route planner **208** and other points to find a valid target waypoint **340** to provide to a path solver **314** that generates a regional path **348**.

[0065] In the example of FIG. 4B, the regional path planner **122** has identified a candidate target waypoint **436** as described herein with respect to FIG. 3. The waypoint validator **338** determines whether the candidate target waypoint **436** is valid. Since the candidate target waypoint **436** is occupied by an unmapped obstacle **438**, the waypoint validator **338** determines that the candidate target waypoint **436** is invalid, and searches for a replacement target waypoint by performing a longitudinal search and/or a lateral search. The longitudinal and lateral searches can start at the invalid target waypoint **436**. Alternatively, the longitudinal and/or lateral searches can start at a point on the global route plan that is past the location of the mobile robot **402** and precedes the invalid target waypoint **436**. The search attempts to find one or more valid points that are closer to the destination global route waypoint **442** than is the current location of the mobile robot **402**. That is, the search attempts to find valid points that enable the mobile robot **402** to make progress toward the destination global route waypoint **442**. The longitudinal search is along the global route plan toward the destination global route waypoint **442**. The lateral search is perpendicular to the global route plan at the invalid target waypoint **436** and is toward the destination global route waypoint **442**, so that progress can be made toward the destination global route waypoint **442**.

[0066] In some embodiments, upon determining that the target waypoint **436** is invalid, the waypoint manager **334** searches for valid points longitudinally and laterally along the global route plan starting from the invalid target waypoint **436**. The longitudinal search proceeds along the global route plan as shown by an initial arrow from the target waypoint **436** to the global route waypoint **404C**, and by a further arrow from the global route waypoint **404C** toward the destination global route waypoint **442**. There are no valid points on the route segment between the target waypoint **436** and the global route waypoint **404C**, since that segment is within the unmapped

obstacle **438**. There are valid points on the route segment between the global route waypoint **404C** and the destination global route waypoint **442**. The valid points are between the unmapped obstacle **438** and the planning horizon **430**. The waypoint manager **334** identifies a valid point shown as replacement target waypoint **444**. The waypoint manager **334** can use the replacement target waypoint **444** as a valid target waypoint **340** to provide to the path solver **314**.

[0067] In some embodiments, upon determining that the target waypoint **436** is invalid, the waypoint manager **334** performs a lateral search for valid points starting from the invalid candidate target waypoint **436**. The lateral search proceeds in a direction perpendicular to the global route plan at the invalid candidate target waypoint **436** as shown by an arrow from the invalid candidate target waypoint **436** to the lateral candidate waypoint **440A**. The lateral candidate waypoint **440A** is a valid point because it is not occupied by an obstacle and is within the planning horizon **430**. The waypoint manager **334** can use the lateral candidate waypoint **440A** as a valid target waypoint **340** to provide to the path solver **314**.

[0068] In some embodiments, upon determining that the target waypoint **436** is invalid, the waypoint manager **334** performs a longitudinal search from the invalid candidate target waypoint **436** to the global route waypoint **404C**, then performs a lateral search from the global route waypoint **404C**, which proceeds in a direction perpendicular to the global route plan at the global route waypoint **404C** as shown by an arrow from the global route waypoint **404C** to the lateral candidate waypoint **440B**. The lateral candidate waypoint **440B** is a valid point because it is not occupied by an obstacle and is within the planning horizon **430**. The waypoint manager **334** can use the lateral candidate waypoint **440A** as a valid target waypoint **340** to provide to the path solver **314**.

[0069] Now referring to FIG. 5, each block of method **500**, described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The method may also be embodied as computer-usable instructions stored on computer storage media. The method may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, method **500** is described, by way of example, with respect to the system of FIGS. 1-3 and 6. However, this method may additionally or alternatively be executed by any one system, or any combination of systems, including, but not limited to, those described herein. Further, the operations in method **500** may be omitted, repeated, and/or performed in any order without departing from the scope of the present disclosure.

[0070] FIG. 5 illustrates a flow diagram of a method **500** for generating a regional path, according to various embodiments. As shown in FIG. 5, method **500** begins with operation **502**, in which a regional path planner **122** receives a global route plan that includes a plurality of waypoints **210** representing locations in a physical environment. In operation **504**, the regional path planner **122** identifies a search space that includes at least a portion of the global route plan. The search space can be the navigable regions **216**, which include at least a portion of the global route waypoints **210** that form the global route plan. In operation **506**, the regional path planner **122** generates a discretized space **318** that represents the search space, where the discretized space **318** includes a plurality of cells representing areas of the search space. The discretized space **318** can be generated by the space discretizer **316** from the navigable regions **216**. The discretized space **318** includes a grid **320A** or lattice **320B** representing areas of the discretized space **318**.

[0071] In operation **508**, the regional path planner **122** searches the discretized space **318** for a target waypoint of the plurality of waypoints. The search starts from a current location of a mobile robot. For example, the waypoint manager **334** can search the global route waypoints **210** for a candidate target waypoint **336**. The candidate target waypoint **336** can be a waypoint in the global route waypoints **210** that is between the current location of the mobile robot and the final destination waypoint of the global route waypoints **210**.

[0072] In operation **510**, the regional path planner **122** determines whether the target waypoint found by the search is invalid. The target waypoint is invalid if, for example, it is occupied (e.g., by an obstacle) or is at a distance from the mobile robot greater than a planning horizon. If operation **510** determines that the target waypoint found by the search is invalid, then in operation **512**, the regional path planner **122** searches the discretized space for a replacement target waypoint. The search starts from the invalid target waypoint. After performing operation **512**, the regional path planner **122** performs operation **510** again to determine whether the replacement target waypoint is invalid.

[0073] If operation **510** determines that the target waypoint found by the search is valid, then the regional path planner **122** performs operation **514**. In operation **514**, the regional path planner **122** searches the search space for a path between the current location of the mobile robot and the target waypoint. For example, a path solver **314** uses a path searcher **344** to search for a regional path **348** to a valid target waypoint **340** via one or more candidate next state(s) **312** provided by a candidate next state generator **310**.

[0074] In operation **516**, the regional path planner **122** determines whether the search performed in operation **514** found a path. If operation **516** determines that the search found the path, then the regional path planner **122** performs operation **518**. At operation **518**, the regional path planner **122** causes the mobile robot to navigate to the target waypoint via the path.

[0075] If operation **516** determines that the search did not find a path, when the regional path planner **122** performs operation **520**. At operation **520**, the regional path planner **122** identifies an expanded search space that includes the search space and is larger than the search space. Subsequent to performing operation **520**, the regional path planner **122** performs operation **514** again to search for another path. Upon determining that operation **516** found another path, the regional path planner **122** performs operation **518** to cause the mobile robot to navigate to the target waypoint via the path.

[0076] FIG. **6** is a more detailed illustration of the speed decision maker **124** of FIG. **1**, according to various embodiments. The speed decision maker **124** generates a set of speed directives **652** that specify how the speed of a mobile robot is to be changed to avoid collisions with obstacles as the mobile robot navigates along a regional path **348**. The speed decision maker **124** generates speed constraints **616** using input that includes a path such as a regional path **348**, a distance map **224**, and/or navigable regions **218** extracted from a semantic map **206**.

[0077] The distance map **224** and/or the navigable regions **218** indicate the locations of obstacles in the physical environment and also include semantic information about obstacles and/or about regions of the physical environment. The navigable regions **218** identify locations of static (e.g., stationary) obstacles and/or speed zones that are associated with speed limits. The distance map **224** identifies the locations of dynamic (e.g., moving or changing) obstacles. The semantic information about obstacles can include a type of each obstacle and a specification of motion of the obstacle. The semantic information about regions can include a speed limit associated with a navigable region **218**, for example.

[0078] The speed decision maker **124** includes a problem formulator **602**, a speed solver **632**, and a speed directive generator **650**. The problem formulator **602** generates one or more speed constraints **616**, which represent a speed problem. The speed solver **632** solves the speed problem by generating one or more speed change decisions **634** for the speed constraints **616**. The speed change decisions **634** specify speed changes that are to occur at specified times during navigation. The speed change decisions **634** represent a speed profile satisfying the speed constraints **616**.

[0079] The speed decision maker **124** uses one or more obstacle reasoners **604** to generate one or more speed constraints **616** based on characteristics of the identified obstacles. Each obstacle reasoner **604** is associated with one or more particular types of obstacle and includes a constraint generator **614** that generates speed constraints **616** for the associated obstacle type(s). Each obstacle reasoner **604** uses the distance map **224** and/or navigable regions **218** to identify one or

more obstacles that are of the obstacle type(s) **606** handled by the respective obstacle reasoner **604** and are on or near a given regional path **348** in the physical environment. An obstacle is near the given regional path **348** if the obstacle is within a threshold distance of the regional path **348**, for example. When an obstacle reasoner **604** that handles obstacles of a particular obstacle type **606** identifies an obstacle of the particular obstacle type **606** in the navigable regions **218** or distance map **224**, and the obstacle location **608** specified for the obstacle in the navigable regions **218** or distance map **224** is on or near the regional path **348**, the obstacle reasoner **604** provides the obstacle location **608** of the obstacle and an obstacle motion description **610** of the obstacle (if present) to a constraint generator **614** that generates speed constraints **616** for obstacles of the particular obstacle type **606**.

[0080] The speed constraints **616** generated by the obstacle reasoners **604** can include a speed zone constraint **618**, a stopping constraint **626**, and/or other constraints that specify limits on speed to be imposed under particular conditions. Since there can be multiple moving obstacles, multiple speed constraints can apply during the same time period and at the same location. A speed solver **632** uses optimization techniques to find a set of speed change decisions **634** that minimize the overall cost of violations of the speed constraints **616**. Accordingly, each speed constraint **616** determines a cost to penalize constraint violations. The cost can be proportional to an amount by which the constraint is violated. A speed zone constraint **618**, for example, determines a cost that penalizes speed limit violations for mobile robot locations that are within a polygon. A speed zone constraint **618** determines a cost that penalizes mobile robot speeds that exceed a speed limit **624** when the mobile robot is within an effective spatial range **620** (e.g., a polygon or a longitudinal distance) during an effective time range **622**. A stopping constraint **626** is similar to a speed zone constraint **618**, but determines a cost that penalizes travel violations if they occur beyond an effective spatial range **628** (e.g., a polygon or a longitudinal distance) during an effective time range **630**. A stopping constraint **626** can correspond to a speed zone constraint **618** having a speed limit **624** of zero.

Since the regional path **348** is given as an input, the locations along the regional path **348** at which speed constraints **616** are to be enforced can be specified as one-dimensional distances, e.g., as a start and an end distance. For dynamic objects, the times during which speed constraints **616** are to be enforced can be specified as ranges of elapsed travel time, e.g., as a start time and an end time.

[0081] As an example, an automatic door reasoner generates a stopping constraint **626** to impose costs on mobile robot travel beyond a door when the door is closed, and generates a speed zone constraint **618** to impose costs on mobile robot travel above a speed limit when the door is open. As another example, a stair reasoner generates a stopping constraint to impose costs on mobile robot travel beyond a stair to avoid falling down the stairs, and a speed zone constraint to impose costs on mobile robot travel above a speed limit within a threshold distance of the stairs. As still another example, a pedestrian reasoner generates a stopping constraint to impose costs on mobile robot travel within a particular threshold distance of pedestrians, and a speed zone constraint to impose costs on mobile robot travel within a farther threshold distance from pedestrians. Multiple obstacle reasoners **604** can operate concurrently. The speed solver **632** can resolve conflicts that arise between speed constraints **616**.

[0082] The speed solver **632** generates a set of speed change decisions **634** that change the speed of the mobile robot over time in accordance with the speed constraints **616**. The speed solver **632** uses optimization techniques to generate the speed change decisions **634**. The set of speed change decisions **634** includes at least one speed change decision **636**. A speed change decision **636** can be generated for each speed constraint **616**, for example. Each speed change decision **636** includes a speed constraint **638** and an action **640**. The speed constraint **638** can identify or reference the corresponding speed constraint **616** from which the speed change decision **636** has been generated. The action **640** can be a stop action, an accelerate action, a decelerate action, or a coast action, for example. The accelerate action causes the speed of the mobile robot to increase to a specified speed. The specified speed can be the speed limit associated with a speed zone constraint or other

specified speed. The decelerate action causes the speed of the mobile robot to decrease to a specified speed. The coast action causes the robot to maintain the current speed for the duration of the planning horizon. The stop action causes the mobile robot to stop moving, e.g., decelerate to a speed of zero.

[0083] The speed solver **632** uses one or more optimization techniques to find a set of speed change decisions **634** that optimize (e.g., minimize) the overall cost of violations of the speed constraints **616**. The speed solver **632** can search a path-time space to identify speed profiles that satisfy the constraints, for example. In some embodiments, the speed solver **632** divides the path-time space into discrete intervals and search for a solution in each interval. In other embodiments, the speed solver **632** performs a speed profile rollout, which involves sampling feasible speed controls of the mobile robot, such as acceleration and jerk, to generate motion primitives and explore the path-time space.

[0084] The speed directive generator **650** converts the speed change decisions **634** to speed directives **652** suitable for input to a local trajectory planner **670**. The speed directives **652** can specify speed limits that apply during particular regions and times. Each speed directive **652** can be a speed zone directive **654**, a stop directive **662**, or other directive specifying constraints on speed. Each speed zone directive **654** includes a speed limit **660**, a spatial range **656** specifying the location in discretized space **318** to which the speed limit **660** applies, and time range **658** during which the speed limit **660** applies. Each stop directive **662** includes a spatial range **664** in which the mobile robot is to stop, and a time range **666** during which the mobile robot is to stop.

[0085] The speed directive generator **650** generates the attributes of each speed directives **652**, such as a spatial range **656**, based on the corresponding attributes of the speed constraints **616** of the respective speed change decisions **634**. The speed directive generator **650** generates a speed zone directive **654** for each speed change decision **636** for which the speed constraint **638** corresponds to a speed zone constraint **618**. For example, to generate a speed zone directive **654** from a speed change decision **636**, the spatial range **656**, time range **658**, and speed limit **660** of the speed zone directive **654** are set to the effective spatial range **620**, the effective time range **622**, and the speed limit **624**, respectively, of the speed zone constraint **618** associated with the speed change decision **636**. As another example, to generate a stop directive **662** from a speed change decision **636**, the spatial range **664** and time range **666** of the stop directive **662** are set to the respective effective spatial range **628** and effective time range **630** of the stopping constraint **626** associated with the speed change decision **636**.

[0086] The speed directives **652** are suitable for use as input to the local trajectory planner **670**. The local trajectory planner **670** generates local trajectories **232** that satisfy the speed directives **652** and thus satisfy the speed constraints **616** in accordance with the speed change decisions **634** generated by the speed solver **632**.

[0087] FIG. 7 illustrates a navigable space **700** that includes static and dynamic obstacles, and a path-time space representation **710** of the navigable space **700**, according to various embodiments. A mobile robot **702** is located on a path **704** in the navigable space **700**. A pedestrian **706** is moving towards the path **704** of the mobile robot **702**. A closed automatic door **708** is blocking the path of the mobile robot **702**. The path-time space **710** of the navigable space **700** illustrates the locations of the pedestrian **706** and closed automatic door **708** on a two-dimensional graph having a path distance axis **712** and a time axis **714**. The path distance axis **712** represents distance along the path **704** and the time axis **714** represents elapsed time since the initial position of the mobile robot **702** shown in the navigable space **700**. A pedestrian region **716** represents the location of the pedestrian **706** along the path **704** in time and space. The pedestrian **706** is located between path distances p_0 and p_1 , and between times t_0 and t_1 . The closed automatic door **708** is located between path distances p_2 and p_3 , and between times 0 and t_2 . The motion of the mobile robot **702** along the path **704** over time is represented by speed profiles **720**, **724**. As shown by the speed profiles **720**, the mobile robot **702** has an initial speed that would cause the speed profile **720** to intersect with

the pedestrian region **716**. Such an intersection between a speed profile and an obstacle region indicates that the mobile robot **702** collides with the obstacle. The speed decision maker **124** generates speed directives **652** that cause the mobile robot **702** to avoid colliding with the pedestrian **706** and automatic door **708**.

[0088] For example, the problem formulator **602** includes a pedestrian obstacle reasoner **604** and a door obstacle reasoner **604**. The pedestrian obstacle reasoner **604** identifies the pedestrian **706** and the obstacle location **608** of the pedestrian **706** in the distance map **224**, and determines the obstacle motion description **610** of the pedestrian **706** based on the motion (e.g., speed) of the pedestrian. The obstacle motion description **610** of the pedestrian **706** indicates that the pedestrian **706** is moving in a particular direction (e.g., to the west) at a particular speed (e.g., 3 km/hour). Accordingly, the pedestrian obstacle reasoner **604** generates a speed zone constraint **618** specifying a speed limit **624** for the mobile robot **702** (e.g., 5 km/hour) within an effective spatial range **620** (e.g., from p0 minus a safety margin through p1 plus a safety margin) and/or within an effective time range **622** (e.g., from t0 minus a safety margin through t1 plus a safety margin). The pedestrian obstacle reasoner **604** also generates a stopping constraint **626** specifying that the mobile robot **702** is to stop within an effective spatial range **620** (e.g., from p0 through p1) and/or within an effective time range **622** (e.g., from t0 through t1).

[0089] As another example, the door obstacle reasoner **604** identifies the automatic door **708** and the obstacle location **608** of the automatic door **708** in a navigable region **216** extracted from a semantic map **206**. The door obstacle reasoner **604** also extracts semantic information describing the motion of the door from the navigable region **216** or semantic map **206**, and determines the obstacle motion description **610** of the automatic door **708** based on the semantic information. The obstacle motion description **610** of the automatic door **708** indicates that the automatic door **708** is closed and blocks the path between times p2 and p3. Accordingly, the pedestrian obstacle reasoner **604** generates a stopping constraint **626** specifying that the mobile robot **702** is to stop within an effective spatial range **620** (e.g., from p2 through p3) and/or an effective time range **622** (e.g., from 0 through t2).

[0090] The speed solver **632** searches the path-time space **710** to identify speed change decisions **634** that avoid the pedestrian region **716** and the automatic door region **718**. The example speed profiles **720**, **724** correspond to speed change decisions **634** generated by the speed solver **632** and/or speed directives **652** generated by the speed directive generator **650** from the speed change decisions **634**. In one example, as shown by the slower speed profile **720**, to avoid collision with the pedestrian region **716** and automatic door region **718**, the speed solver **632** can generate a speed change decision **634** specifying a decelerate action **722** to be performed at a time prior to t0 and/or at path distance prior to p0. By decelerating from the initial speed, the slower speed profile **720** causes the mobile robot **702** to arrive at the location on the path **704** crossed by the pedestrian **706** after the pedestrian **706** has crossed the path. In this way, the mobile robot **702** avoids a collision with the pedestrian **706**.

[0091] In another example, as shown by the faster speed profile **724**, to avoid collision, the speed solver **632** can generate a speed change decision **634** specifying an accelerate action **726** to be performed at a time prior to t0 and/or at a path distance prior to p0. By accelerating from the initial speed, the faster speed profile **724** causes the mobile robot **702** to arrive at the location on the path **704** crossed by the pedestrian **706** before the pedestrian **706** crosses the path, thereby avoiding a collision with the pedestrian **706**. The speed resulting from the accelerate action **726** is sufficiently high that the mobile robot **702** will collide with the automatic door region **718** if the speed is not reduced. Thus, the speed solver **632** generates a decelerate action **730**, which is performed at a time prior to t0 as shown in the path-time space **710**. The decelerate action **730** slows the speed of the mobile robot **702** sufficiently so that the mobile robot **702** does not reach the automatic door **708** during the time the automatic door **708** is closed. In this way, the mobile robot **702** avoids a collision with the automatic door **708**.

[0092] Now referring to FIG. 8, each block of method **800**, described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory. The method may also be embodied as computer-usable instructions stored on computer storage media. The method may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, method **800** is described, by way of example, with respect to the system of FIGS. **1-3** and **6**. However, this method may additionally or alternatively be executed by any one system, or any combination of systems, including, but not limited to, those described herein. Further, the operations in method **800** can be omitted, repeated, and/or performed in any order without departing from the scope of the present disclosure.

[0093] FIG. **8** illustrates a flow diagram of a method for generating speed directives **228** for a path, according to various embodiments. As shown in FIG. **8**, method **800** begins with operation **802**, in which a speed decision maker **124** identifies, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment. For example, the maps can include a semantic map **206** received from a mapping system **204** and/or a distance map **224** received from a perception system **220**. The semantic map **206** and/or the distance map **224** indicate the locations of obstacles in the physical environment and also can include semantic information about obstacles and/or about regions of the physical environment.

[0094] The speed decision maker **124** includes one or more obstacle reasoners **604**. Each obstacle reasoner **604** is associated with one or more obstacle type(s) **606** for which the obstacle reasoner **604** handles processing tasks as described herein. Each obstacle reasoner **604** uses the distance map **224** and/or navigable regions **218** to identify one or more obstacles that are of the obstacle type(s) **606** handled by the respective obstacle reasoner **604** and are on or near a given regional path **348** in the physical environment. An obstacle is near the given regional path **348** if the obstacle is within a threshold distance of the regional path **348**, for example.

[0095] In operation **804**, the speed decision maker **124** generates, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path. For example, when an obstacle reasoner **604** that handles obstacles of a particular obstacle type **606** identifies an obstacle of the particular obstacle type **606** in the navigable regions **218** or distance map **224**, and the obstacle location **608** specified for the obstacle in the navigable regions **218** or distance map **224** is on or near the regional path **348**, the obstacle reasoner **604** provides the obstacle location **608** of the obstacle and an obstacle motion description **610** of the obstacle (if present) to a constraint generator **614** that generates speed constraints **616** for obstacles of the particular obstacle type **606**. The speed constraints **616** generated by the obstacle reasoners **604** can include a speed zone constraint **618**, a stopping constraint **626**, and/or other constraints that specify limits on speed to be imposed under particular conditions.

[0096] In operation **806**, the speed decision maker **124** generates one or more speed change decisions that determine one or more speeds of the robot over time, the determined speeds of the robot over time satisfying the speed constraints. For example, the speed decision maker **124** includes a speed solver **632** that uses optimization techniques to find a set of speed change decisions **634** minimizing the overall cost of violations of the speed constraints **616**. Each speed constraint **616** determines a cost to penalize constraint violations. The speed solver **632** generates a set of speed change decisions **634** that change the speed of the mobile robot over time in accordance with the speed constraints **616**. For example, the speed change decisions **634** can be generated by the speed solver **632** such that the sum of the costs determined by the speed change decisions **634** for a speed profile corresponding to the speed change decisions **634** is optimized (e.g., minimized relative to other candidate sets of speed change decisions **634** identified by the speed solver **632**).

[0097] In operation **808**, the speed decision maker **124** converts the speed change decisions **634** to one or more speed directives **228** specifying speed limits for the mobile robot in specified distance ranges during specified time ranges. For example, the speed decision maker **124** can include a speed directive generator **650** that converts the speed change decisions **634** to corresponding speed directives **228**.

[0098] In operation **810**, the speed decision maker **124** causes the mobile robot to navigate in accordance with the speed directives **228**. The speed directives **228** are suitable for input to a local trajectory planner **670**, and the speed decision maker **124** provides the speed directives **228** to the local trajectory planner **230**. The local trajectory planner **230** also receives a path, such as a regional path **226**, and generates a local trajectory **232** using the speed directives **228** and the regional path **226**. The local trajectory planner **230** provides the local trajectory **232** to a robot controller **234**, which causes the robot to navigate in accordance with the speed directives **228** and the regional path **226**.

[0099] The systems and methods described herein may be used by, without limitation, non-autonomous vehicles or machines, semi-autonomous vehicles or machines (e.g., in one or more adaptive driver assistance systems (ADAS)), autonomous vehicles or machines, piloted and unpiloted robots or robotic platforms, warehouse vehicles, off-road vehicles, vehicles coupled to one or more trailers, flying vessels, boats, shuttles, emergency response vehicles, motorcycles, electric or motorized bicycles, aircraft, construction vehicles, underwater craft, drones, and/or other vehicle types. Further, the systems and methods described herein may be used for a variety of purposes, by way of example and without limitation, for machine control, machine locomotion, machine driving, synthetic data generation, model training, perception, augmented reality, virtual reality, mixed reality, robotics, security and surveillance, simulation and digital twinning, autonomous or semi-autonomous machine applications, deep learning, environment simulation, object or actor simulation and/or digital twinning, data center processing, conversational AI, light transport simulation (e.g., ray-tracing, path tracing, etc.), collaborative content creation for 3D assets, cloud computing, and/or any other suitable applications.

[0100] Disclosed embodiments may be comprised in a variety of different systems such as automotive systems (e.g., a control system for an autonomous or semi-autonomous machine, a perception system for an autonomous or semi-autonomous machine), systems implemented using a robot, aerial systems, medial systems, boating systems, smart area monitoring systems, systems for performing deep learning operations, systems for performing simulation operations, systems for performing digital twin operations, systems implemented using an edge device, systems incorporating one or more virtual machines (VMs), systems for performing synthetic data generation operations, systems implemented at least partially in a data center, systems for performing conversational AI operations, systems implementing one or more language models—such as large language models (LLMs) that process text, audio, and/or sensor data, systems for performing light transport simulation, systems for performing collaborative content creation for 3D assets, systems for performing generative AI operations, systems implemented at least partially using cloud computing resources, and/or other types of systems.

Example Autonomous Vehicle

[0101] FIG. **91A** is an illustration of an example autonomous vehicle **9100**, in accordance with some embodiments of the present disclosure. The autonomous vehicle **9100** (alternatively referred to herein as the “vehicle **9100**”) may include, without limitation, a passenger vehicle, such as a car, a truck, a bus, a first responder vehicle, a shuttle, an electric or motorized bicycle, a motorcycle, a fire truck, a police vehicle, an ambulance, a boat, a construction vehicle, an underwater craft, a robotic vehicle, a drone, an airplane, a vehicle coupled to a trailer (e.g., a semi-tractor-trailer truck used for hauling cargo), and/or another type of vehicle (e.g., that is unmanned and/or that accommodates one or more passengers). Autonomous vehicles are generally described in terms of automation levels, defined by the National Highway Traffic Safety Administration (NHTSA), a

division of the US Department of Transportation, and the Society of Automotive Engineers (SAE) “Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles” (Standard No. J3016-201806, published on Jun. 15, 2018, Standard No. J3016-201609, published on Sep. 30, 2016, and previous and future versions of this standard). The vehicle **9100** may be capable of functionality in accordance with one or more of Level 3-Level 5 of the autonomous driving levels. The vehicle **9100** may be capable of functionality in accordance with one or more of Level 1-Level 5 of the autonomous driving levels. For example, the vehicle **9100** may be capable of driver assistance (Level 1), partial automation (Level 2), conditional automation (Level 3), high automation (Level 4), and/or full automation (Level 5), depending on the embodiment. The term “autonomous,” as used herein, may include any and/or all types of autonomy for the vehicle **9100** or other machine, such as being fully autonomous, being highly autonomous, being conditionally autonomous, being partially autonomous, providing assistive autonomy, being semi-autonomous, being primarily autonomous, or other designation.

[0102] The vehicle **9100** may include components such as a chassis, a vehicle body, wheels (e.g., 2, 4, 6, 8, 18, etc.), tires, axles, and other components of a vehicle. The vehicle **9100** may include a propulsion system **9150**, such as an internal combustion engine, hybrid electric power plant, an all-electric engine, and/or another propulsion system type. The propulsion system **9150** may be connected to a drive train of the vehicle **9100**, which may include a transmission, to enable the propulsion of the vehicle **9100**. The propulsion system **9150** may be controlled in response to receiving signals from the throttle/accelerator **9152**.

[0103] A steering system **9154**, which may include a steering wheel, may be used to steer the vehicle **9100** (e.g., along a desired path or route) when the propulsion system **9150** is operating (e.g., when the vehicle is in motion). The steering system **9154** may receive signals from a steering actuator **9156**. The steering wheel may be optional for full automation (Level 5) functionality.

[0104] The brake sensor system **9146** may be used to operate the vehicle brakes in response to receiving signals from the brake actuators **9148** and/or brake sensors.

[0105] Controller(s) **9136**, which may include one or more system on chips (SoCs) **9104** (FIG. **91C**) and/or GPU(s), may provide signals (e.g., representative of commands) to one or more components and/or systems of the vehicle **9100**. For example, the controller(s) may send signals to operate the vehicle brakes via one or more brake actuators **9148**, to operate the steering system **9154** via one or more steering actuators **9156**, to operate the propulsion system **9150** via one or more throttle/accelerators **9152**. The controller(s) **9136** may include one or more onboard (e.g., integrated) computing devices (e.g., supercomputers) that process sensor signals, and output operation commands (e.g., signals representing commands) to enable autonomous driving and/or to assist a human driver in driving the vehicle **9100**. The controller(s) **9136** may include a first controller **9136** for autonomous driving functions, a second controller **9136** for functional safety functions, a third controller **9136** for artificial intelligence functionality (e.g., computer vision), a fourth controller **9136** for infotainment functionality, a fifth controller **9136** for redundancy in emergency conditions, and/or other controllers. In some examples, a single controller **9136** may handle two or more of the above functionalities, two or more controllers **9136** may handle a single functionality, and/or any combination thereof.

[0106] The controller(s) **9136** may provide the signals for controlling one or more components and/or systems of the vehicle **9100** in response to sensor data received from one or more sensors (e.g., sensor inputs). The sensor data may be received from, for example and without limitation, global navigation satellite systems (“GNSS”) sensor(s) **9158** (e.g., Global Positioning System sensor(s)), RADAR sensor(s) **9160**, ultrasonic sensor(s) **9162**, LIDAR sensor(s) **9164**, inertial measurement unit (IMU) sensor(s) **9166** (e.g., accelerometer(s), gyroscope(s), magnetic compass(es), magnetometer(s), etc.), microphone(s) **9196**, stereo camera(s) **9168**, wide-view camera(s) **9170** (e.g., fisheye cameras), infrared camera(s) **9172**, surround camera(s) **9174** (e.g., 360 degree cameras), long-range and/or mid-range camera(s) **9198**, speed sensor(s) **9144** (e.g., for

measuring the speed of the vehicle **9100**), vibration sensor(s) **9142**, steering sensor(s) **9140**, brake sensor(s) (e.g., as part of the brake sensor system **9146**), and/or other sensor types.

[0107] One or more of the controller(s) **9136** may receive inputs (e.g., represented by input data) from an instrument cluster **9132** of the vehicle **9100** and provide outputs (e.g., represented by output data, display data, etc.) via a human-machine interface (HMI) display **9134**, an audible annunciator, a loudspeaker, and/or via other components of the vehicle **9100**. The outputs may include information such as vehicle velocity, speed, time, map data (e.g., the High Definition (“HD”) map **9122** of FIG. **91C**), location data (e.g., the vehicle's **9100** location, such as on a map), direction, location of other vehicles (e.g., an occupancy grid), information about objects and status of objects as perceived by the controller(s) **9136**, etc. For example, the HMI display **9134** may display information about the presence of one or more objects (e.g., a street sign, caution sign, traffic light changing, etc.), and/or information about driving maneuvers the vehicle has made, is making, or will make (e.g., changing lanes now, taking exit **34B** in two miles, etc.).

[0108] The vehicle **9100** further includes a network interface **9124** which may use one or more wireless antenna(s) **9126** and/or modem(s) to communicate over one or more networks. For example, the network interface **9124** may be capable of communication over Long-Term Evolution (“LTE”), Wideband Code Division Multiple Access (“WCDMA”), Universal Mobile Telecommunications System (“UMTS”), Global System for Mobile communication (“GSM”), IMT-CDMA Multi-Carrier (“CDMA2000”), etc. The wireless antenna(s) **9126** may also enable communication between objects in the environment (e.g., vehicles, mobile devices, etc.), using local area network(s), such as Bluetooth, Bluetooth Low Energy (“LE”), Z-Wave, ZigBee, etc., and/or low power wide-area network(s) (“LPWANs”), such as LoRaWAN, SigFox, etc.

[0109] FIG. **91B** is an example of camera locations and fields of view for the example autonomous vehicle **9100** of FIG. **91A**, in accordance with some embodiments of the present disclosure. The cameras and respective fields of view are one example embodiment and are not intended to be limiting. For example, additional and/or alternative cameras may be included and/or the cameras may be located at different locations on the vehicle **9100**.

[0110] The camera types for the cameras may include, but are not limited to, digital cameras that may be adapted for use with the components and/or systems of the vehicle **9100**. The camera(s) may operate at automotive safety integrity level (ASIL) B and/or at another ASIL. The camera types may be capable of any image capture rate, such as 60 frames per second (fps), 120 fps, 240 fps, etc., depending on the embodiment. The cameras may be capable of using rolling shutters, global shutters, another type of shutter, or a combination thereof. In some examples, the color filter array may include a red clear clear clear (RCCC) color filter array, a red clear clear blue (RCCB) color filter array, a red blue green clear (RBGC) color filter array, a Foveon X3 color filter array, a Bayer sensors (RGGB) color filter array, a monochrome sensor color filter array, and/or another type of color filter array. In some embodiments, clear pixel cameras, such as cameras with an RCCC, an RCCB, and/or an RBGC color filter array, may be used in an effort to increase light sensitivity.

[0111] In some examples, one or more of the camera(s) may be used to perform advanced driver assistance systems (ADAS) functions (e.g., as part of a redundant or fail-safe design). For example, a Multi-Function Mono Camera may be installed to provide functions including lane departure warning, traffic sign assist and intelligent headlamp control. One or more of the camera(s) (e.g., all of the cameras) may record and provide image data (e.g., video) simultaneously.

[0112] One or more of the cameras may be mounted in a mounting assembly, such as a custom designed (three dimensional (“3D”) printed) assembly, in order to cut out stray light and reflections from within the car (e.g., reflections from the dashboard reflected in the windshield mirrors) which may interfere with the camera's image data capture abilities. With reference to wing-mirror mounting assemblies, the wing-mirror assemblies may be custom 3D printed so that the camera mounting plate matches the shape of the wing-mirror. In some examples, the camera(s) may be

integrated into the wing-mirror. For side-view cameras, the camera(s) may also be integrated within the four pillars at each corner of the cabin.

[0113] Cameras with a field of view that include portions of the environment in front of the vehicle **9100** (e.g., front-facing cameras) may be used for surround view, to help identify forward facing paths and obstacles, as well aid in, with the help of one or more controllers **9136** and/or control SoCs, providing information critical to generating an occupancy grid and/or determining the preferred vehicle paths. Front-facing cameras may be used to perform many of the same ADAS functions as LIDAR, including emergency braking, pedestrian detection, and collision avoidance. Front-facing cameras may also be used for ADAS functions and systems including Lane Departure Warnings (“LDW”), Autonomous Cruise Control (“ACC”), and/or other functions such as traffic sign recognition.

[0114] A variety of cameras may be used in a front-facing configuration, including, for example, a monocular camera platform that includes a complementary metal oxide semiconductor (“CMOS”) color imager. Another example may be a wide-view camera(s) **9170** that may be used to perceive objects coming into view from the periphery (e.g., pedestrians, crossing traffic or bicycles). Although only one wide-view camera is illustrated in FIG. **91B**, there may be any number (including zero) of wide-view cameras **9170** on the vehicle **9100**. In addition, any number of long-range camera(s) **9198** (e.g., a long-view stereo camera pair) may be used for depth-based object detection, especially for objects for which a neural network has not yet been trained. The long-range camera(s) **9198** may also be used for object detection and classification, as well as basic object tracking.

[0115] Any number of stereo cameras **9168** may also be included in a front-facing configuration. In at least one embodiment, one or more of stereo camera(s) **9168** may include an integrated control unit comprising a scalable processing unit, which may provide a programmable logic (“FPGA”) and a multi-core micro-processor with an integrated Controller Area Network (“CAN”) or Ethernet interface on a single chip. Such a unit may be used to generate a 3D map of the vehicle's environment, including a distance estimate for all the points in the image. An alternative stereo camera(s) **9168** may include a compact stereo vision sensor(s) that may include two camera lenses (one each on the left and right) and an image processing chip that may measure the distance from the vehicle to the target object and use the generated information (e.g., metadata) to activate the autonomous emergency braking and lane departure warning functions. Other types of stereo camera(s) **9168** may be used in addition to, or alternatively from, those described herein.

[0116] Cameras with a field of view that include portions of the environment to the side of the vehicle **9100** (e.g., side-view cameras) may be used for surround view, providing information used to create and update the occupancy grid, as well as to generate side impact collision warnings. For example, surround camera(s) **9174** (e.g., four surround cameras **9174** as illustrated in FIG. **91B**) may be positioned to on the vehicle **9100**. The surround camera(s) **9174** may include wide-view camera(s) **9170**, fisheye camera(s), 360 degree camera(s), and/or the like. Four example, four fisheye cameras may be positioned on the vehicle's front, rear, and sides. In an alternative arrangement, the vehicle may use three surround camera(s) **9174** (e.g., left, right, and rear), and may leverage one or more other camera(s) (e.g., a forward-facing camera) as a fourth surround view camera.

[0117] Cameras with a field of view that include portions of the environment to the rear of the vehicle **9100** (e.g., rear-view cameras) may be used for park assistance, surround view, rear collision warnings, and creating and updating the occupancy grid. A wide variety of cameras may be used including, but not limited to, cameras that are also suitable as a front-facing camera(s) (e.g., long-range and/or mid-range camera(s) **9198**, stereo camera(s) **9168**), infrared camera(s) **9172**, etc.), as described herein.

[0118] FIG. **91C** is a block diagram of an example system architecture for the example autonomous vehicle **9100** of FIG. **91A**, in accordance with some embodiments of the present disclosure. It

should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.

[0119] Each of the components, features, and systems of the vehicle **9100** in FIG. **91C** are illustrated as being connected via bus **9102**. The bus **9102** may include a Controller Area Network (CAN) data interface (alternatively referred to herein as a “CAN bus”). A CAN may be a network inside the vehicle **9100** used to aid in control of various features and functionality of the vehicle **9100**, such as actuation of brakes, acceleration, braking, steering, windshield wipers, etc. A CAN bus may be configured to have dozens or even hundreds of nodes, each with its own unique identifier (e.g., a CAN ID). The CAN bus may be read to find steering wheel angle, ground speed, engine revolutions per minute (RPMs), button positions, and/or other vehicle status indicators. The CAN bus may be ASIL B compliant.

[0120] Although the bus **9102** is described herein as being a CAN bus, this is not intended to be limiting. For example, in addition to, or alternatively from, the CAN bus, FlexRay and/or Ethernet may be used. Additionally, although a single line is used to represent the bus **9102**, this is not intended to be limiting. For example, there may be any number of busses **9102**, which may include one or more CAN busses, one or more FlexRay busses, one or more Ethernet busses, and/or one or more other types of busses using a different protocol. In some examples, two or more busses **9102** may be used to perform different functions, and/or may be used for redundancy. For example, a first bus **9102** may be used for collision avoidance functionality and a second bus **9102** may be used for actuation control. In any example, each bus **9102** may communicate with any of the components of the vehicle **9100**, and two or more busses **9102** may communicate with the same components. In some examples, each SoC **9104**, each controller **9136**, and/or each computer within the vehicle may have access to the same input data (e.g., inputs from sensors of the vehicle **9100**), and may be connected to a common bus, such the CAN bus.

[0121] The vehicle **9100** may include one or more controller(s) **9136**, such as those described herein with respect to FIG. **91A**. The controller(s) **9136** may be used for a variety of functions. The controller(s) **9136** may be coupled to any of the various other components and systems of the vehicle **9100**, and may be used for control of the vehicle **9100**, artificial intelligence of the vehicle **9100**, infotainment for the vehicle **9100**, and/or the like.

[0122] The vehicle **9100** may include a system(s) on a chip (SoC) **9104**. The SoC **9104** may include CPU(s) **9106**, GPU(s) **9108**, processor(s) **9110**, cache(s) **9112**, accelerator(s) **9114**, data store(s) **9116**, and/or other components and features not illustrated. The SoC(s) **9104** may be used to control the vehicle **9100** in a variety of platforms and systems. For example, the SoC(s) **9104** may be combined in a system (e.g., the system of the vehicle **9100**) with an HD map **9122** which may obtain map refreshes and/or updates via a network interface **9124** from one or more servers (e.g., server(s) **9178** of FIG. **91D**).

[0123] The CPU(s) **9106** may include a CPU cluster or CPU complex (alternatively referred to herein as a “CCPLEX”). The CPU(s) **9106** may include multiple cores and/or L2 caches. For example, in some embodiments, the CPU(s) **9106** may include eight cores in a coherent multi-processor configuration. In some embodiments, the CPU(s) **9106** may include four dual-core clusters where each cluster has a dedicated L2 cache (e.g., a 2 MB L2 cache). The CPU(s) **9106** (e.g., the CPLEX) may be configured to support simultaneous cluster operation enabling any combination of the clusters of the CPU(s) **9106** to be active at any given time.

[0124] The CPU(s) **9106** may implement power management capabilities that include one or more of the following features: individual hardware blocks may be clock-gated automatically when idle to save dynamic power; each core clock may be gated when the core is not actively executing instructions due to execution of WFI/WFE instructions; each core may be independently power-gated; each core cluster may be independently clock-gated when all cores are clock-gated or power-gated; and/or each core cluster may be independently power-gated when all cores are power-gated. The CPU(s) **9106** may further implement an enhanced algorithm for managing power states, where allowed power states and expected wakeup times are specified, and the hardware/microcode determines the best power state to enter for the core, cluster, and CCPLX. The processing cores may support simplified power state entry sequences in software with the work offloaded to microcode.

[0125] The GPU(s) **9108** may include an integrated GPU (alternatively referred to herein as an “iGPU”). The GPU(s) **9108** may be programmable and may be efficient for parallel workloads. The GPU(s) **9108**, in some examples, may use an enhanced tensor instruction set. The GPU(s) **9108** may include one or more streaming microprocessors, where each streaming microprocessor may include an L1 cache (e.g., an L1 cache with at least 96 KB storage capacity), and two or more of the streaming microprocessors may share an L2 cache (e.g., an L2 cache with a 512 KB storage capacity). In some embodiments, the GPU(s) **9108** may include at least eight streaming microprocessors. The GPU(s) **9108** may use compute application programming interface(s) (API(s)). In addition, the GPU(s) **9108** may use one or more parallel computing platforms and/or programming models (e.g., NVIDIA's CUDA).

[0126] The GPU(s) **9108** may be power-optimized for best performance in automotive and embedded use cases. For example, the GPU(s) **9108** may be fabricated on a Fin field-effect transistor (FinFET). However, this is not intended to be limiting and the GPU(s) **9108** may be fabricated using other semiconductor manufacturing processes. Each streaming microprocessor may incorporate a number of mixed-precision processing cores partitioned into multiple blocks. For example, and without limitation, 64 PF32 cores and 32 PF64 cores may be partitioned into four processing blocks. In such an example, each processing block may be allocated 16 FP32 cores, 8 FP64 cores, 16 INT32 cores, two mixed-precision NVIDIA TENSOR COREs for deep learning matrix arithmetic, an L0 instruction cache, a warp scheduler, a dispatch unit, and/or a 64 KB register file. In addition, the streaming microprocessors may include independent parallel integer and floating-point data paths to provide for efficient execution of workloads with a mix of computation and addressing calculations. The streaming microprocessors may include independent thread scheduling capability to enable finer-grain synchronization and cooperation between parallel threads. The streaming microprocessors may include a combined L1 data cache and shared memory unit in order to improve performance while simplifying programming.

[0127] The GPU(s) **9108** may include a high bandwidth memory (HBM) and/or a 16 GB HBM2 memory subsystem to provide, in some examples, about 900 GB/second peak memory bandwidth. In some examples, in addition to, or alternatively from, the HBM memory, a synchronous graphics random-access memory (SGRAM) may be used, such as a graphics double data rate type five synchronous random-access memory (GDDR5).

[0128] The GPU(s) **9108** may include unified memory technology including access counters to allow for more accurate migration of memory pages to the processor that accesses them most frequently, thereby improving efficiency for memory ranges shared between processors. In some examples, address translation services (ATS) support may be used to allow the GPU(s) **9108** to access the CPU(s) **9106** page tables directly. In such examples, when the GPU(s) **9108** memory management unit (MMU) experiences a miss, an address translation request may be transmitted to the CPU(s) **9106**. In response, the CPU(s) **9106** may look in its page tables for the virtual-to-physical mapping for the address and transmits the translation back to the GPU(s) **9108**. As such, unified memory technology may allow a single unified virtual address space for memory of both

the CPU(s) **9106** and the GPU(s) **9108**, thereby simplifying the GPU(s) **9108** programming and porting of applications to the GPU(s) **9108**.

[0129] In addition, the GPU(s) **9108** may include an access counter that may keep track of the frequency of access of the GPU(s) **9108** to memory of other processors. The access counter may help ensure that memory pages are moved to the physical memory of the processor that is accessing the pages most frequently.

[0130] The SoC(s) **9104** may include any number of cache(s) **9112**, including those described herein. For example, the cache(s) **9112** may include an L3 cache that is available to both the CPU(s) **9106** and the GPU(s) **9108** (e.g., that is connected both the CPU(s) **9106** and the GPU(s) **9108**). The cache(s) **9112** may include a write-back cache that may keep track of states of lines, such as by using a cache coherence protocol (e.g., MEI, MESI, MSI, etc.). The L3 cache may include 4 MB or more, depending on the embodiment, although smaller cache sizes may be used.

[0131] The SoC(s) **9104** may include an arithmetic logic unit(s) (ALU(s)) which may be leveraged in performing processing with respect to any of the variety of tasks or operations of the vehicle **9100**—such as processing DNNs. In addition, the SoC(s) **9104** may include a floating point unit(s) (FPU(s))—or other math coprocessor or numeric coprocessor types—for performing mathematical operations within the system. For example, the SoC(s) **104** may include one or more FPUs integrated as execution units within a CPU(s) **9106** and/or GPU(s) **9108**.

[0132] The SoC(s) **9104** may include one or more accelerators **9114** (e.g., hardware accelerators, software accelerators, or a combination thereof). For example, the SoC(s) **9104** may include a hardware acceleration cluster that may include optimized hardware accelerators and/or large on-chip memory. The large on-chip memory (e.g., 4 MB of SRAM), may enable the hardware acceleration cluster to accelerate neural networks and other calculations. The hardware acceleration cluster may be used to complement the GPU(s) **9108** and to off-load some of the tasks of the GPU(s) **9108** (e.g., to free up more cycles of the GPU(s) **9108** for performing other tasks). As an example, the accelerator(s) **9114** may be used for targeted workloads (e.g., perception, convolutional neural networks (CNNs), etc.) that are stable enough to be amenable to acceleration. The term “CNN,” as used herein, may include all types of CNNs, including region-based or regional convolutional neural networks (RCNNs) and Fast RCNNs (e.g., as used for object detection).

[0133] The accelerator(s) **9114** (e.g., the hardware acceleration cluster) may include a deep learning accelerator(s) (DLA). The DLA(s) may include one or more Tensor processing units (TPUs) that may be configured to provide an additional ten trillion operations per second for deep learning applications and inferencing. The TPUs may be accelerators configured to, and optimized for, performing image processing functions (e.g., for CNNs, RCNNs, etc.). The DLA(s) may further be optimized for a specific set of neural network types and floating point operations, as well as inferencing. The design of the DLA(s) may provide more performance per millimeter than a general-purpose GPU, and vastly exceeds the performance of a CPU. The TPU(s) may perform several functions, including a single-instance convolution function, supporting, for example, INT8, INT16, and FP16 data types for both features and weights, as well as post-processor functions.

[0134] The DLA(s) may quickly and efficiently execute neural networks, especially CNNs, on processed or unprocessed data for any of a variety of functions, including, for example and without limitation: a CNN for object identification and detection using data from camera sensors; a CNN for distance estimation using data from camera sensors; a CNN for emergency vehicle detection and identification and detection using data from microphones; a CNN for facial recognition and vehicle owner identification using data from camera sensors; and/or a CNN for security and/or safety related events.

[0135] The DLA(s) may perform any function of the GPU(s) **9108**, and by using an inference accelerator, for example, a designer may target either the DLA(s) or the GPU(s) **9108** for any function. For example, the designer may focus processing of CNNs and floating point operations

on the DLA(s) and leave other functions to the GPU(s) **9108** and/or other accelerator(s) **9114**.

[0136] The accelerator(s) **9114** (e.g., the hardware acceleration cluster) may include a programmable vision accelerator(s) (PVA), which may alternatively be referred to herein as a computer vision accelerator. The PVA(s) may be designed and configured to accelerate computer vision algorithms for the advanced driver assistance systems (ADAS), autonomous driving, and/or augmented reality (AR) and/or virtual reality (VR) applications. The PVA(s) may provide a balance between performance and flexibility. For example, each PVA(s) may include, for example and without limitation, any number of reduced instruction set computer (RISC) cores, direct memory access (DMA), and/or any number of vector processors.

[0137] The RISC cores may interact with image sensors (e.g., the image sensors of any of the cameras described herein), image signal processor(s), and/or the like. Each of the RISC cores may include any amount of memory. The RISC cores may use any of a number of protocols, depending on the embodiment. In some examples, the RISC cores may execute a real-time operating system (RTOS). The RISC cores may be implemented using one or more integrated circuit devices, application specific integrated circuits (ASICs), and/or memory devices. For example, the RISC cores may include an instruction cache and/or a tightly coupled RAM.

[0138] The DMA may enable components of the PVA(s) to access the system memory independently of the CPU(s) **9106**. The DMA may support any number of features used to provide optimization to the PVA including, but not limited to, supporting multi-dimensional addressing and/or circular addressing. In some examples, the DMA may support up to six or more dimensions of addressing, which may include block width, block height, block depth, horizontal block stepping, vertical block stepping, and/or depth stepping.

[0139] The vector processors may be programmable processors that may be designed to efficiently and flexibly execute programming for computer vision algorithms and provide signal processing capabilities. In some examples, the PVA may include a PVA core and two vector processing subsystem partitions. The PVA core may include a processor subsystem, DMA engine(s) (e.g., two DMA engines), and/or other peripherals. The vector processing subsystem may operate as the primary processing engine of the PVA, and may include a vector processing unit (VPU), an instruction cache, and/or vector memory (e.g., VMEM). A VPU core may include a digital signal processor such as, for example, a single instruction, multiple data (SIMD), very long instruction word (VLIW) digital signal processor. The combination of the SIMD and VLIW may enhance throughput and speed.

[0140] Each of the vector processors may include an instruction cache and may be coupled to dedicated memory. As a result, in some examples, each of the vector processors may be configured to execute independently of the other vector processors. In other examples, the vector processors that are included in a particular PVA may be configured to employ data parallelism. For example, in some embodiments, the plurality of vector processors included in a single PVA may execute the same computer vision algorithm, but on different regions of an image. In other examples, the vector processors included in a particular PVA may simultaneously execute different computer vision algorithms, on the same image, or even execute different algorithms on sequential images or portions of an image. Among other things, any number of PVAs may be included in the hardware acceleration cluster and any number of vector processors may be included in each of the PVAs. In addition, the PVA(s) may include additional error correcting code (ECC) memory, to enhance overall system safety.

[0141] The accelerator(s) **9114** (e.g., the hardware acceleration cluster) may include a computer vision network on-chip and SRAM, for providing a high-bandwidth, low latency SRAM for the accelerator(s) **9114**. In some examples, the on-chip memory may include at least 4 MB SRAM, consisting of, for example and without limitation, eight field-configurable memory blocks, that may be accessible by both the PVA and the DLA. Each pair of memory blocks may include an advanced peripheral bus (APB) interface, configuration circuitry, a controller, and a multiplexer.

Any type of memory may be used. The PVA and DLA may access the memory via a backbone that provides the PVA and DLA with high-speed access to memory. The backbone may include a computer vision network on-chip that interconnects the PVA and the DLA to the memory (e.g., using the APB).

[0142] The computer vision network on-chip may include an interface that determines, before transmission of any control signal/address/data, that both the PVA and the DLA provide ready and valid signals. Such an interface may provide for separate phases and separate channels for transmitting control signals/addresses/data, as well as burst-type communications for continuous data transfer. This type of interface may comply with ISO 26262 or IEC 61508 standards, although other standards and protocols may be used.

[0143] In some examples, the SoC(s) **9104** may include a real-time ray-tracing hardware accelerator, such as described in U.S. patent application Ser. No. 16/101,232, filed on Aug. 10, 2018. The real-time ray-tracing hardware accelerator may be used to quickly and efficiently determine the positions and extents of objects (e.g., within a world model), to generate real-time visualization simulations, for RADAR signal interpretation, for sound propagation synthesis and/or analysis, for simulation of SONAR systems, for general wave propagation simulation, for comparison to LIDAR data for purposes of localization and/or other functions, and/or for other uses. In some embodiments, one or more tree traversal units (TTUs) may be used for executing one or more ray-tracing related operations.

[0144] The accelerator(s) **9114** (e.g., the hardware accelerator cluster) have a wide array of uses for autonomous driving. The PVA may be a programmable vision accelerator that may be used for key processing stages in ADAS and autonomous vehicles. The PVA's capabilities are a good match for algorithmic domains needing predictable processing, at low power and low latency. In other words, the PVA performs well on semi-dense or dense regular computation, even on small data sets, which need predictable run-times with low latency and low power. Thus, in the context of platforms for autonomous vehicles, the PVAs are designed to run classic computer vision algorithms, as they are efficient at object detection and operating on integer math.

[0145] For example, according to one embodiment of the technology, the PVA is used to perform computer stereo vision. A semi-global matching-based algorithm may be used in some examples, although this is not intended to be limiting. Many applications for Level 3-5 autonomous driving require motion estimation/stereo matching on-the-fly (e.g., structure from motion, pedestrian recognition, lane detection, etc.). The PVA may perform computer stereo vision function on inputs from two monocular cameras.

[0146] In some examples, the PVA may be used to perform dense optical flow. According to process raw RADAR data (e.g., using a 4D Fast Fourier Transform) to provide Processed RADAR. In other examples, the PVA is used for time of flight depth processing, by processing raw time of flight data to provide processed time of flight data, for example.

[0147] The DLA may be used to run any type of network to enhance control and driving safety, including for example, a neural network that outputs a measure of confidence for each object detection. Such a confidence value may be interpreted as a probability, or as providing a relative “weight” of each detection compared to other detections. This confidence value enables the system to make further decisions regarding which detections should be considered as true positive detections rather than false positive detections. For example, the system may set a threshold value for the confidence and consider only the detections exceeding the threshold value as true positive detections. In an automatic emergency braking (AEB) system, false positive detections would cause the vehicle to automatically perform emergency braking, which is obviously undesirable. Therefore, only the most confident detections should be considered as triggers for AEB. The DLA may run a neural network for regressing the confidence value. The neural network may take as its input at least some subset of parameters, such as bounding box dimensions, ground plane estimate obtained (e.g. from another subsystem), inertial measurement unit (IMU) sensor **9166** output that

correlates with the vehicle **9100** orientation, distance, 3D location estimates of the object obtained from the neural network and/or other sensors (e.g., LIDAR sensor(s) **9164** or RADAR sensor(s) **9160**), among others.

[0148] The SoC(s) **9104** may include data store(s) **9116** (e.g., memory). The data store(s) **9116** may be on-chip memory of the SoC(s) **9104**, which may store neural networks to be executed on the GPU and/or the DLA. In some examples, the data store(s) **9116** may be large enough in capacity to store multiple instances of neural networks for redundancy and safety. The data store(s) **9112** may comprise L2 or L3 cache(s) **9112**. Reference to the data store(s) **9116** may include reference to the memory associated with the PVA, DLA, and/or other accelerator(s) **9114**, as described herein.

[0149] The SoC(s) **9104** may include one or more processor(s) **9110** (e.g., embedded processors). The processor(s) **9110** may include a boot and power management processor that may be a dedicated processor and subsystem to handle boot power and management functions and related security enforcement. The boot and power management processor may be a part of the SoC(s) **9104** boot sequence and may provide runtime power management services. The boot power and management processor may provide clock and voltage programming, assistance in system low power state transitions, management of SoC(s) **9104** thermals and temperature sensors, and/or management of the SoC(s) **9104** power states. Each temperature sensor may be implemented as a ring-oscillator whose output frequency is proportional to temperature, and the SoC(s) **9104** may use the ring-oscillators to detect temperatures of the CPU(s) **9106**, GPU(s) **9108**, and/or accelerator(s) **9114**. If temperatures are determined to exceed a threshold, the boot and power management processor may enter a temperature fault routine and put the SoC(s) **9104** into a lower power state and/or put the vehicle **9100** into a chauffeur to safe stop mode (e.g., bring the vehicle **9100** to a safe stop).

[0150] The processor(s) **9110** may further include a set of embedded processors that may serve as an audio processing engine. The audio processing engine may be an audio subsystem that enables full hardware support for multi-channel audio over multiple interfaces, and a broad and flexible range of audio I/O interfaces. In some examples, the audio processing engine is a dedicated processor core with a digital signal processor with dedicated RAM.

[0151] The processor(s) **9110** may further include an always on processor engine that may provide necessary hardware features to support low power sensor management and wake use cases. The always on processor engine may include a processor core, a tightly coupled RAM, supporting peripherals (e.g., timers and interrupt controllers), various I/O controller peripherals, and routing logic.

[0152] The processor(s) **9110** may further include a safety cluster engine that includes a dedicated processor subsystem to handle safety management for automotive applications. The safety cluster engine may include two or more processor cores, a tightly coupled RAM, support peripherals (e.g., timers, an interrupt controller, etc.), and/or routing logic. In a safety mode, the two or more cores may operate in a lockstep mode and function as a single core with comparison logic to detect any differences between their operations.

[0153] The processor(s) **9110** may further include a real-time camera engine that may include a dedicated processor subsystem for handling real-time camera management.

[0154] The processor(s) **9110** may further include a high-dynamic range signal processor that may include an image signal processor that is a hardware engine that is part of the camera processing pipeline.

[0155] The processor(s) **9110** may include a video image compositor that may be a processing block (e.g., implemented on a microprocessor) that implements video post-processing functions needed by a video playback application to produce the final image for the player window. The video image compositor may perform lens distortion correction on wide-view camera(s) **9170**, surround camera(s) **9174**, and/or on in-cabin monitoring camera sensors. In-cabin monitoring camera sensor is preferably monitored by a neural network running on another instance of the

Advanced SoC, configured to identify in-cabin events and respond accordingly. An in-cabin system may perform lip reading to activate cellular service and place a phone call, dictate emails, change the vehicle's destination, activate or change the vehicle's infotainment system and settings, or provide voice-activated web surfing. Certain functions are available to the driver only when the vehicle is operating in an autonomous mode, and are disabled otherwise.

[0156] The video image compositor may include enhanced temporal noise reduction for both spatial and temporal noise reduction. For example, where motion occurs in a video, the noise reduction weights spatial information appropriately, decreasing the weight of information provided by adjacent frames. Where an image or portion of an image does not include motion, the temporal noise reduction performed by the video image compositor may use information from the previous image to reduce noise in the current image.

[0157] The video image compositor may also be configured to perform stereo rectification on input stereo lens frames. The video image compositor may further be used for user interface composition when the operating system desktop is in use, and the GPU(s) **9108** is not required to continuously render new surfaces. Even when the GPU(s) **9108** is powered on and active doing 3D rendering, the video image compositor may be used to offload the GPU(s) **9108** to improve performance and responsiveness.

[0158] The SoC(s) **9104** may further include a mobile industry processor interface (MIPI) camera serial interface for receiving video and input from cameras, a high-speed interface, and/or a video input block that may be used for camera and related pixel input functions. The SoC(s) **9104** may further include an input/output controller(s) that may be controlled by software and may be used for receiving I/O signals that are uncommitted to a specific role.

[0159] The SoC(s) **9104** may further include a broad range of peripheral interfaces to enable communication with peripherals, audio codecs, power management, and/or other devices. The SoC(s) **9104** may be used to process data from cameras (e.g., connected over Gigabit Multimedia Serial Link and Ethernet), sensors (e.g., LIDAR sensor(s) **9164**, RADAR sensor(s) **9160**, etc. that may be connected over Ethernet), data from bus **9102** (e.g., speed of vehicle **9100**, steering wheel position, etc.), data from GNSS sensor(s) **9158** (e.g., connected over Ethernet or CAN bus). The SoC(s) **9104** may further include dedicated high-performance mass storage controllers that may include their own DMA engines, and that may be used to free the CPU(s) **9106** from routine data management tasks.

[0160] The SoC(s) **9104** may be an end-to-end platform with a flexible architecture that spans automation levels 3-5, thereby providing a comprehensive functional safety architecture that leverages and makes efficient use of computer vision and ADAS techniques for diversity and redundancy, provides a platform for a flexible, reliable driving software stack, along with deep learning tools. The SoC(s) **9104** may be faster, more reliable, and even more energy-efficient and space-efficient than conventional systems. For example, the accelerator(s) **9114**, when combined with the CPU(s) **9106**, the GPU(s) **9108**, and the data store(s) **9116**, may provide for a fast, efficient platform for level 3-5 autonomous vehicles.

[0161] The technology thus provides capabilities and functionality that cannot be achieved by conventional systems. For example, computer vision algorithms may be executed on CPUs, which may be configured using high-level programming language, such as the C programming language, to execute a wide variety of processing algorithms across a wide variety of visual data. However, CPUs are oftentimes unable to meet the performance requirements of many computer vision applications, such as those related to execution time and power consumption, for example. In particular, many CPUs are unable to execute complex object detection algorithms in real-time, which is a requirement of in-vehicle ADAS applications, and a requirement for practical Level 3-5 autonomous vehicles.

[0162] In contrast to conventional systems, by providing a CPU complex, GPU complex, and a hardware acceleration cluster, the technology described herein allows for multiple neural networks

to be performed simultaneously and/or sequentially, and for the results to be combined together to enable Level 3-5 autonomous driving functionality. For example, a CNN executing on the DLA or dGPU (e.g., the GPU(s) **9120**) may include a text and word recognition, allowing the supercomputer to read and understand traffic signs, including signs for which the neural network has not been specifically trained. The DLA may further include a neural network that is able to identify, interpret, and provides semantic understanding of the sign, and to pass that semantic understanding to the path planning modules running on the CPU Complex.

[0163] As another example, multiple neural networks may be run simultaneously, as is required for Level 3, 4, or 5 driving. For example, a warning sign consisting of “Caution: flashing lights indicate icy conditions,” along with an electric light, may be independently or collectively interpreted by several neural networks. The sign itself may be identified as a traffic sign by a first deployed neural network (e.g., a neural network that has been trained), the text “Flashing lights indicate icy conditions” may be interpreted by a second deployed neural network, which informs the vehicle's path planning software (preferably executing on the CPU Complex) that when flashing lights are detected, icy conditions exist. The flashing light may be identified by operating a third deployed neural network over multiple frames, informing the vehicle's path-planning software of the presence (or absence) of flashing lights. All three neural networks may run simultaneously, such as within the DLA and/or on the GPU(s) **9108**.

[0164] In some examples, a CNN for facial recognition and vehicle owner identification may use data from camera sensors to identify the presence of an authorized driver and/or owner of the vehicle **9100**. The always on sensor processing engine may be used to unlock the vehicle when the owner approaches the driver door and turn on the lights, and, in security mode, to disable the vehicle when the owner leaves the vehicle. In this way, the SoC(s) **9104** provide for security against theft and/or carjacking.

[0165] In another example, a CNN for emergency vehicle detection and identification may use data from microphones **9196** to detect and identify emergency vehicle sirens. In contrast to conventional systems, that use general classifiers to detect sirens and manually extract features, the SoC(s) **9104** use the CNN for classifying environmental and urban sounds, as well as classifying visual data. In a preferred embodiment, the CNN running on the DLA is trained to identify the relative closing speed of the emergency vehicle (e.g., by using the Doppler Effect). The CNN may also be trained to identify emergency vehicles specific to the local area in which the vehicle is operating, as identified by GNSS sensor(s) **9158**. Thus, for example, when operating in Europe the CNN will seek to detect European sirens, and when in the United States the CNN will seek to identify only North American sirens. Once an emergency vehicle is detected, a control program may be used to execute an emergency vehicle safety routine, slowing the vehicle, pulling over to the side of the road, parking the vehicle, and/or idling the vehicle, with the assistance of ultrasonic sensors **9162**, until the emergency vehicle(s) passes.

[0166] The vehicle may include a CPU(s) **9118** (e.g., discrete CPU(s), or dCPU(s)), that may be coupled to the SoC(s) **9104** via a high-speed interconnect (e.g., PCIe). The CPU(s) **9118** may include an X86 processor, for example. The CPU(s) **9118** may be used to perform any of a variety of functions, including arbitrating potentially inconsistent results between ADAS sensors and the SoC(s) **9104**, and/or monitoring the status and health of the controller(s) **9136** and/or infotainment SoC **9130**, for example.

[0167] The vehicle **9100** may include a GPU(s) **9120** (e.g., discrete GPU(s), or dGPU(s)), that may be coupled to the SoC(s) **9104** via a high-speed interconnect (e.g., NVIDIA's NVLINK). The GPU(s) **9120** may provide additional artificial intelligence functionality, such as by executing redundant and/or different neural networks, and may be used to train and/or update neural networks based on input (e.g., sensor data) from sensors of the vehicle **9100**.

[0168] The vehicle **9100** may further include the network interface **9124** which may include one or more wireless antennas **9126** (e.g., one or more wireless antennas for different communication

protocols, such as a cellular antenna, a Bluetooth antenna, etc.). The network interface **9124** may be used to enable wireless connectivity over the Internet with the cloud (e.g., with the server(s) **9178** and/or other network devices), with other vehicles, and/or with computing devices (e.g., client devices of passengers). To communicate with other vehicles, a direct link may be established between the two vehicles and/or an indirect link may be established (e.g., across networks and over the Internet). Direct links may be provided using a vehicle-to-vehicle communication link. The vehicle-to-vehicle communication link may provide the vehicle **9100** information about vehicles in proximity to the vehicle **9100** (e.g., vehicles in front of, on the side of, and/or behind the vehicle **9100**). This functionality may be part of a cooperative adaptive cruise control functionality of the vehicle **9100**.

[0169] The network interface **9124** may include a SoC that provides modulation and demodulation functionality and enables the controller(s) **9136** to communicate over wireless networks. The network interface **9124** may include a radio frequency front-end for up-conversion from baseband to radio frequency, and down conversion from radio frequency to baseband. The frequency conversions may be performed through well-known processes, and/or may be performed using super-heterodyne processes. In some examples, the radio frequency front end functionality may be provided by a separate chip. The network interface may include wireless functionality for communicating over LTE, WCDMA, UMTS, GSM, CDMA2000, Bluetooth, Bluetooth LE, Wi-Fi, Z-Wave, ZigBee, LoRaWAN, and/or other wireless protocols.

[0170] The vehicle **9100** may further include data store(s) **9128** which may include off-chip (e.g., off the SoC(s) **9104**) storage. The data store(s) **9128** may include one or more storage elements including RAM, SRAM, DRAM, VRAM, Flash, hard disks, and/or other components and/or devices that may store at least one bit of data.

[0171] The vehicle **9100** may further include GNSS sensor(s) **9158**. The GNSS sensor(s) **9158** (e.g., GPS, assisted GPS sensors, differential GPS (DGPS) sensors, etc.), to assist in mapping, perception, occupancy grid generation, and/or path planning functions. Any number of GNSS sensor(s) **9158** may be used, including, for example and without limitation, a GPS using a USB connector with an Ethernet to Serial (RS-232) bridge.

[0172] The vehicle **9100** may further include RADAR sensor(s) **9160**. The RADAR sensor(s) **9160** may be used by the vehicle **9100** for long-range vehicle detection, even in darkness and/or severe weather conditions. RADAR functional safety levels may be ASIL B. The RADAR sensor(s) **9160** may use the CAN and/or the bus **9102** (e.g., to transmit data generated by the RADAR sensor(s) **9160**) for control and to access object tracking data, with access to Ethernet to access raw data in some examples. A wide variety of RADAR sensor types may be used. For example, and without limitation, the RADAR sensor(s) **9160** may be suitable for front, rear, and side RADAR use. In some example, Pulse Doppler RADAR sensor(s) are used.

[0173] The RADAR sensor(s) **9160** may include different configurations, such as long range with narrow field of view, short range with wide field of view, short range side coverage, etc. In some examples, long-range RADAR may be used for adaptive cruise control functionality. The long-range RADAR systems may provide a broad field of view realized by two or more independent scans, such as within a 250 m range. The RADAR sensor(s) **9160** may help in distinguishing between static and moving objects, and may be used by ADAS systems for emergency brake assist and forward collision warning. Long-range RADAR sensors may include monostatic multimodal RADAR with multiple (e.g., six or more) fixed RADAR antennae and a high-speed CAN and FlexRay interface. In an example with six antennae, the central four antennae may create a focused beam pattern, designed to record the vehicle's **9100** surroundings at higher speeds with minimal interference from traffic in adjacent lanes. The other two antennae may expand the field of view, making it possible to quickly detect vehicles entering or leaving the vehicle's **9100** lane.

[0174] Mid-range RADAR systems may include, as an example, a range of up to 9160 m (front) or 80 m (rear), and a field of view of up to 42 degrees (front) or 9150 degrees (rear). Short-range

RADAR systems may include, without limitation, RADAR sensors designed to be installed at both ends of the rear bumper. When installed at both ends of the rear bumper, such a RADAR sensor systems may create two beams that constantly monitor the blind spot in the rear and next to the vehicle.

[0175] Short-range RADAR systems may be used in an ADAS system for blind spot detection and/or lane change assist.

[0176] The vehicle **9100** may further include ultrasonic sensor(s) **9162**. The ultrasonic sensor(s) **9162**, which may be positioned at the front, back, and/or the sides of the vehicle **9100**, may be used for park assist and/or to create and update an occupancy grid. A wide variety of ultrasonic sensor(s) **9162** may be used, and different ultrasonic sensor(s) **9162** may be used for different ranges of detection (e.g., 2.5 m, 4 m). The ultrasonic sensor(s) **9162** may operate at functional safety levels of ASIL B.

[0177] The vehicle **9100** may include LIDAR sensor(s) **9164**. The LIDAR sensor(s) **9164** may be used for object and pedestrian detection, emergency braking, collision avoidance, and/or other functions. The LIDAR sensor(s) **9164** may be functional safety level ASIL B. In some examples, the vehicle **9100** may include multiple LIDAR sensors **9164** (e.g., two, four, six, etc.) that may use Ethernet (e.g., to provide data to a Gigabit Ethernet switch).

[0178] In some examples, the LIDAR sensor(s) **9164** may be capable of providing a list of objects and their distances for a 360-degree field of view. Commercially available LIDAR sensor(s) **9164** may have an advertised range of approximately 9100 m, with an accuracy of 2 cm-3 cm, and with support for a 9100 Mbps Ethernet connection, for example. In some examples, one or more non-protruding LIDAR sensors **9164** may be used. In such examples, the LIDAR sensor(s) **9164** may be implemented as a small device that may be embedded into the front, rear, sides, and/or corners of the vehicle **9100**. The LIDAR sensor(s) **9164**, in such examples, may provide up to a 120-degree horizontal and 35-degree vertical field-of-view, with a 200 m range even for low-reflectivity objects. Front-mounted LIDAR sensor(s) **9164** may be configured for a horizontal field of view between 45 degrees and 135 degrees.

[0179] In some examples, LIDAR technologies, such as 3D flash LIDAR, may also be used. 3D Flash LIDAR uses a flash of a laser as a transmission source, to illuminate vehicle surroundings up to approximately 200 m. A flash LIDAR unit includes a receptor, which records the laser pulse transit time and the reflected light on each pixel, which in turn corresponds to the range from the vehicle to the objects. Flash LIDAR may allow for highly accurate and distortion-free images of the surroundings to be generated with every laser flash. In some examples, four flash LIDAR sensors may be deployed, one at each side of the vehicle **9100**. Available 3D flash LIDAR systems include a solid-state 3D staring array LIDAR camera with no moving parts other than a fan (e.g., a non-scanning LIDAR device). The flash LIDAR device may use a 5 nanosecond class I (eye-safe) laser pulse per frame and may capture the reflected laser light in the form of 3D range point clouds and co-registered intensity data. By using flash LIDAR, and because flash LIDAR is a solid-state device with no moving parts, the LIDAR sensor(s) **9164** may be less susceptible to motion blur, vibration, and/or shock.

[0180] The vehicle may further include IMU sensor(s) **9166**. The IMU sensor(s) **9166** may be located at a center of the rear axle of the vehicle **9100**, in some examples. The IMU sensor(s) **9166** may include, for example and without limitation, an accelerometer(s), a magnetometer(s), a gyroscope(s), a magnetic compass(es), and/or other sensor types. In some examples, such as in six-axis applications, the IMU sensor(s) **9166** may include accelerometers and gyroscopes, while in nine-axis applications, the IMU sensor(s) **9166** may include accelerometers, gyroscopes, and magnetometers.

[0181] In some embodiments, the IMU sensor(s) **9166** may be implemented as a miniature, high performance GPS-Aided Inertial Navigation System (GPS/INS) that combines micro-electro-mechanical systems (MEMS) inertial sensors, a high-sensitivity GPS receiver, and advanced

Kalman filtering algorithms to provide estimates of position, velocity, and attitude. As such, in some examples, the IMU sensor(s) **9166** may enable the vehicle **9100** to estimate heading without requiring input from a magnetic sensor by directly observing and correlating the changes in velocity from GPS to the IMU sensor(s) **9166**. In some examples, the IMU sensor(s) **9166** and the GNSS sensor(s) **9158** may be combined in a single integrated unit.

[0182] The vehicle may include microphone(s) **9196** placed in and/or around the vehicle **9100**. The microphone(s) **9196** may be used for emergency vehicle detection and identification, among other things.

[0183] The vehicle may further include any number of camera types, including stereo camera(s) **9168**, wide-view camera(s) **9170**, infrared camera(s) **9172**, surround camera(s) **9174**, long-range and/or mid-range camera(s) **9198**, and/or other camera types. The cameras may be used to capture image data around an entire periphery of the vehicle **9100**. The types of cameras used depends on the embodiments and requirements for the vehicle **9100**, and any combination of camera types may be used to provide the necessary coverage around the vehicle **9100**. In addition, the number of cameras may differ depending on the embodiment. For example, the vehicle may include six cameras, seven cameras, ten cameras, twelve cameras, and/or another number of cameras. The cameras may support, as an example and without limitation, Gigabit Multimedia Serial Link (GMSL) and/or Gigabit Ethernet. Each of the camera(s) is described with more detail herein with respect to FIG. **91A** and FIG. **91B**.

[0184] The vehicle **9100** may further include vibration sensor(s) **9142**. The vibration sensor(s) **9142** may measure vibrations of components of the vehicle, such as the axle(s). For example, changes in vibrations may indicate a change in road surfaces. In another example, when two or more vibration sensors **9142** are used, the differences between the vibrations may be used to determine friction or slippage of the road surface (e.g., when the difference in vibration is between a power-driven axle and a freely rotating axle).

[0185] The vehicle **9100** may include an ADAS system **9138**. The ADAS system **9138** may include a SoC, in some examples. The ADAS system **9138** may include autonomous/adaptive/automatic cruise control (ACC), cooperative adaptive cruise control (CACC), forward crash warning (FCW), automatic emergency braking (AEB), lane departure warnings (LDW), lane keep assist (LKA), blind spot warning (BSW), rear cross-traffic warning (RCTW), collision warning systems (CWS), lane centering (LC), and/or other features and functionality.

[0186] The ACC systems may use RADAR sensor(s) **9160**, LIDAR sensor(s) **9164**, and/or a camera(s). The ACC systems may include longitudinal ACC and/or lateral ACC. Longitudinal ACC monitors and controls the distance to the vehicle immediately ahead of the vehicle **9100** and automatically adjust the vehicle speed to maintain a safe distance from vehicles ahead. Lateral ACC performs distance keeping, and advises the vehicle **9100** to change lanes when necessary. Lateral ACC is related to other ADAS applications such as LCA and CWS.

[0187] CACC uses information from other vehicles that may be received via the network interface **9124** and/or the wireless antenna(s) **9126** from other vehicles via a wireless link, or indirectly, over a network connection (e.g., over the Internet). Direct links may be provided by a vehicle-to-vehicle (V2V) communication link, while indirect links may be infrastructure-to-vehicle (I2V) communication link. In general, the V2V communication concept provides information about the immediately preceding vehicles (e.g., vehicles immediately ahead of and in the same lane as the vehicle **9100**), while the I2V communication concept provides information about traffic further ahead. CACC systems may include either or both I2V and V2V information sources. Given the information of the vehicles ahead of the vehicle **9100**, CACC may be more reliable and it has potential to improve traffic flow smoothness and reduce congestion on the road.

[0188] FCW systems are designed to alert the driver to a hazard, so that the driver may take corrective action. FCW systems use a front-facing camera and/or RADAR sensor(s) **9160**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback,

such as a display, speaker, and/or vibrating component. FCW systems may provide a warning, such as in the form of a sound, visual warning, vibration and/or a quick brake pulse.

[0189] AEB systems detect an impending forward collision with another vehicle or other object, and may automatically apply the brakes if the driver does not take corrective action within a specified time or distance parameter. AEB systems may use front-facing camera(s) and/or RADAR sensor(s) **9160**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC. When the AEB system detects a hazard, it typically first alerts the driver to take corrective action to avoid the collision and, if the driver does not take corrective action, the AEB system may automatically apply the brakes in an effort to prevent, or at least mitigate, the impact of the predicted collision. AEB systems, may include techniques such as dynamic brake support and/or crash imminent braking.

[0190] LDW systems provide visual, audible, and/or tactile warnings, such as steering wheel or seat vibrations, to alert the driver when the vehicle **9100** crosses lane markings. A LDW system does not activate when the driver indicates an intentional lane departure, by activating a turn signal. LDW systems may use front-side facing cameras, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

[0191] LKA systems are a variation of LDW systems. LKA systems provide steering input or braking to correct the vehicle **9100** if the vehicle **9100** starts to exit the lane.

[0192] BSW systems detects and warn the driver of vehicles in an automobile's blind spot. BSW systems may provide a visual, audible, and/or tactile alert to indicate that merging or changing lanes is unsafe. The system may provide an additional warning when the driver uses a turn signal. BSW systems may use rear-side facing camera(s) and/or RADAR sensor(s) **9160**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

[0193] RCTW systems may provide visual, audible, and/or tactile notification when an object is detected outside the rear-camera range when the vehicle **9100** is backing up. Some RCTW systems include AEB to ensure that the vehicle brakes are applied to avoid a crash. RCTW systems may use one or more rear-facing RADAR sensor(s) **9160**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

[0194] Conventional ADAS systems may be prone to false positive results which may be annoying and distracting to a driver, but typically are not catastrophic, because the ADAS systems alert the driver and allow the driver to decide whether a safety condition truly exists and act accordingly. However, in an autonomous vehicle **9100**, the vehicle **9100** itself must, in the case of conflicting results, decide whether to heed the result from a primary computer or a secondary computer (e.g., a first controller **9136** or a second controller **9136**). For example, in some embodiments, the ADAS system **9138** may be a backup and/or secondary computer for providing perception information to a backup computer rationality module. The backup computer rationality monitor may run a redundant diverse software on hardware components to detect faults in perception and dynamic driving tasks. Outputs from the ADAS system **9138** may be provided to a supervisory MCU. If outputs from the primary computer and the secondary computer conflict, the supervisory MCU must determine how to reconcile the conflict to ensure safe operation.

[0195] In some examples, the primary computer may be configured to provide the supervisory MCU with a confidence score, indicating the primary computer's confidence in the chosen result. If the confidence score exceeds a threshold, the supervisory MCU may follow the primary computer's direction, regardless of whether the secondary computer provides a conflicting or inconsistent result. Where the confidence score does not meet the threshold, and where the primary and secondary computer indicate different results (e.g., the conflict), the supervisory MCU may arbitrate between the computers to determine the appropriate outcome.

[0196] The supervisory MCU may be configured to run a neural network(s) that is trained and

configured to determine, based on outputs from the primary computer and the secondary computer, conditions under which the secondary computer provides false alarms. Thus, the neural network(s) in the supervisory MCU may learn when the secondary computer's output may be trusted, and when it cannot. For example, when the secondary computer is a RADAR-based FCW system, a neural network(s) in the supervisory MCU may learn when the FCW system is identifying metallic objects that are not, in fact, hazards, such as a drainage grate or manhole cover that triggers an alarm. Similarly, when the secondary computer is a camera-based LDW system, a neural network in the supervisory MCU may learn to override the LDW when bicyclists or pedestrians are present and a lane departure is, in fact, the safest maneuver. In embodiments that include a neural network(s) running on the supervisory MCU, the supervisory MCU may include at least one of a DLA or GPU suitable for running the neural network(s) with associated memory. In preferred embodiments, the supervisory MCU may comprise and/or be included as a component of the SoC(s) **9104**.

[0197] In other examples, ADAS system **9138** may include a secondary computer that performs ADAS functionality using traditional rules of computer vision. As such, the secondary computer may use classic computer vision rules (if-then), and the presence of a neural network(s) in the supervisory MCU may improve reliability, safety and performance. For example, the diverse implementation and intentional non-identity makes the overall system more fault-tolerant, especially to faults caused by software (or software-hardware interface) functionality. For example, if there is a software bug or error in the software running on the primary computer, and the non-identical software code running on the secondary computer provides the same overall result, the supervisory MCU may have greater confidence that the overall result is correct, and the bug in software or hardware on primary computer is not causing material error.

[0198] In some examples, the output of the ADAS system **9138** may be fed into the primary computer's perception block and/or the primary computer's dynamic driving task block. For example, if the ADAS system **9138** indicates a forward crash warning due to an object immediately ahead, the perception block may use this information when identifying objects. In other examples, the secondary computer may have its own neural network which is trained and thus reduces the risk of false positives, as described herein.

[0199] The vehicle **9100** may further include the infotainment SoC **9130** (e.g., an in-vehicle infotainment system (IVI)). Although illustrated and described as a SoC, the infotainment system may not be a SoC, and may include two or more discrete components. The infotainment SoC **9130** may include a combination of hardware and software that may be used to provide audio (e.g., music, a personal digital assistant, navigational instructions, news, radio, etc.), video (e.g., TV, movies, streaming, etc.), phone (e.g., hands-free calling), network connectivity (e.g., LTE, Wi-Fi, etc.), and/or information services (e.g., navigation systems, rear-parking assistance, a radio data system, vehicle related information such as fuel level, total distance covered, brake fuel level, oil level, door open/close, air filter information, etc.) to the vehicle **9100**. For example, the infotainment SoC **9130** may radios, disk players, navigation systems, video players, USB and Bluetooth connectivity, carputers, in-car entertainment, Wi-Fi, steering wheel audio controls, hands free voice control, a heads-up display (HUD), an HMI display **9134**, a telematics device, a control panel (e.g., for controlling and/or interacting with various components, features, and/or systems), and/or other components. The infotainment SoC **9130** may further be used to provide information (e.g., visual and/or audible) to a user(s) of the vehicle, such as information from the ADAS system **9138**, autonomous driving information such as planned vehicle maneuvers, trajectories, surrounding environment information (e.g., intersection information, vehicle information, road information, etc.), and/or other information.

[0200] The infotainment SoC **9130** may include GPU functionality. The infotainment SoC **9130** may communicate over the bus **9102** (e.g., CAN bus, Ethernet, etc.) with other devices, systems, and/or components of the vehicle **9100**. In some examples, the infotainment SoC **9130** may be

coupled to a supervisory MCU such that the GPU of the infotainment system may perform some self-driving functions in the event that the primary controller(s) **9136** (e.g., the primary and/or backup computers of the vehicle **9100**) fail. In such an example, the infotainment SoC **9130** may put the vehicle **9100** into a chauffeur to safe stop mode, as described herein.

[0201] The vehicle **9100** may further include an instrument cluster **9132** (e.g., a digital dash, an electronic instrument cluster, a digital instrument panel, etc.). The instrument cluster **9132** may include a controller and/or supercomputer (e.g., a discrete controller or supercomputer). The instrument cluster **9132** may include a set of instrumentation such as a speedometer, fuel level, oil pressure, tachometer, odometer, turn indicators, gearshift position indicator, seat belt warning light(s), parking-brake warning light(s), engine-malfunction light(s), airbag (SRS) system information, lighting controls, safety system controls, navigation information, etc. In some examples, information may be displayed and/or shared among the infotainment SoC **9130** and the instrument cluster **9132**. In other words, the instrument cluster **9132** may be included as part of the infotainment SoC **9130**, or vice versa.

[0202] FIG. **91D** is a system diagram for communication between cloud-based server(s) and the example autonomous vehicle **9100** of FIG. **91A**, in accordance with some embodiments of the present disclosure. The system **9176** may include server(s) **9178**, network(s) **9190**, and vehicles, including the vehicle **9100**. The server(s) **9178** may include a plurality of GPUs **9184(A)-9184(H)** (collectively referred to herein as GPUs **9184**), PCIe switches **9182(A)-9182(H)** (collectively referred to herein as PCIe switches **9182**), and/or CPUs **9180(A)-9180(B)** (collectively referred to herein as CPUs **9180**). The GPUs **9184**, the CPUs **9180**, and the PCIe switches may be interconnected with high-speed interconnects such as, for example and without limitation, NVLink interfaces **9188** developed by NVIDIA and/or PCIe connections **9186**. In some examples, the GPUs **9184** are connected via NVLink and/or NVSwitch SoC and the GPUs **9184** and the PCIe switches **9182** are connected via PCIe interconnects. Although eight GPUs **9184**, two CPUs **9180**, and two PCIe switches are illustrated, this is not intended to be limiting. Depending on the embodiment, each of the server(s) **9178** may include any number of GPUs **9184**, CPUs **9180**, and/or PCIe switches. For example, the server(s) **9178** may each include eight, sixteen, thirty-two, and/or more GPUs **9184**.

[0203] The server(s) **9178** may receive, over the network(s) **9190** and from the vehicles, image data representative of images showing unexpected or changed road conditions, such as recently commenced road-work. The server(s) **9178** may transmit, over the network(s) **9190** and to the vehicles, neural networks **9192**, updated neural networks **9192**, and/or map information **9194**, including information regarding traffic and road conditions. The updates to the map information **9194** may include updates for the HD map **9122**, such as information regarding construction sites, potholes, detours, flooding, and/or other obstructions. In some examples, the neural networks **9192**, the updated neural networks **9192**, and/or the map information **9194** may have resulted from new training and/or experiences represented in data received from any number of vehicles in the environment, and/or based on training performed at a datacenter (e.g., using the server(s) **9178** and/or other servers).

[0204] The server(s) **9178** may be used to train machine learning models (e.g., neural networks) based on training data. The training data may be generated by the vehicles, and/or may be generated in a simulation (e.g., using a game engine). In some examples, the training data is tagged (e.g., where the neural network benefits from supervised learning) and/or undergoes other pre-processing, while in other examples the training data is not tagged and/or pre-processed (e.g., where the neural network does not require supervised learning). Training may be executed according to any one or more classes of machine learning techniques, including, without limitation, classes such as: supervised training, semi-supervised training, unsupervised training, self-learning, reinforcement learning, federated learning, transfer learning, feature learning (including principal component and cluster analyses), multi-linear subspace learning, manifold learning, representation

learning (including sparse dictionary learning), rule-based machine learning, anomaly detection, and any variants or combinations thereof. Once the machine learning models are trained, the machine learning models may be used by the vehicles (e.g., transmitted to the vehicles over the network(s) **9190**, and/or the machine learning models may be used by the server(s) **9178** to remotely monitor the vehicles.

[0205] In some examples, the server(s) **9178** may receive data from the vehicles and apply the data to up-to-date real-time neural networks for real-time intelligent inferencing. The server(s) **9178** may include deep-learning supercomputers and/or dedicated AI computers powered by GPU(s) **9184**, such as a DGX and DGX Station machines developed by NVIDIA. However, in some examples, the server(s) **9178** may include deep learning infrastructure that use only CPU-powered datacenters.

[0206] The deep-learning infrastructure of the server(s) **9178** may be capable of fast, real-time inferencing, and may use that capability to evaluate and verify the health of the processors, software, and/or associated hardware in the vehicle **9100**. For example, the deep-learning infrastructure may receive periodic updates from the vehicle **9100**, such as a sequence of images and/or objects that the vehicle **9100** has located in that sequence of images (e.g., via computer vision and/or other machine learning object classification techniques). The deep-learning infrastructure may run its own neural network to identify the objects and compare them with the objects identified by the vehicle **9100** and, if the results do not match and the infrastructure concludes that the AI in the vehicle **9100** is malfunctioning, the server(s) **9178** may transmit a signal to the vehicle **9100** instructing a fail-safe computer of the vehicle **9100** to assume control, notify the passengers, and complete a safe parking maneuver.

[0207] For inferencing, the server(s) **9178** may include the GPU(s) **9184** and one or more programmable inference accelerators (e.g., NVIDIA's TensorRT). The combination of GPU-powered servers and inference acceleration may make real-time responsiveness possible. In other examples, such as where performance is less critical, servers powered by CPUs, FPGAs, and other processors may be used for inferencing.

Example Computing Device

[0208] FIG. **10** is a block diagram of an example computing device(s) **1000** suitable for use in implementing some embodiments of the present disclosure. Computing device **1000** may include an interconnect system **1002** that directly or indirectly couples the following devices: memory **1004**, one or more central processing units (CPUs) **1006**, one or more graphics processing units (GPUs) **1008**, a communication interface **1010**, input/output (I/O) ports **1012**, input/output components **1014**, a power supply **1016**, one or more presentation components **1018** (e.g., display(s)), and one or more logic units **1020**. In at least one embodiment, the computing device(s) **1000** may comprise one or more virtual machines (VMs), and/or any of the components thereof may comprise virtual components (e.g., virtual hardware components). For non-limiting examples, one or more of the GPUs **1008** may comprise one or more vGPUs, one or more of the CPUs **1006** may comprise one or more vCPUs, and/or one or more of the logic units **1020** may comprise one or more virtual logic units. As such, a computing device(s) **1000** may include discrete components (e.g., a full GPU dedicated to the computing device **1000**), virtual components (e.g., a portion of a GPU dedicated to the computing device **1000**), or a combination thereof.

[0209] Although the various blocks of FIG. **10** are shown as connected via the interconnect system **1002** with lines, this is not intended to be limiting and is for clarity only. For example, in some embodiments, a presentation component **1018**, such as a display device, may be considered an I/O component **1014** (e.g., if the display is a touch screen). As another example, the CPUs **1006** and/or GPUs **1008** may include memory (e.g., the memory **1004** may be representative of a storage device in addition to the memory of the GPUs **1008**, the CPUs **1006**, and/or other components). In other words, the computing device of FIG. **10** is merely illustrative. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “desktop,” “tablet,” “client device,” “mobile

device,” “hand-held device,” “game console,” “electronic control unit (ECU),” “virtual reality system,” and/or other device or system types, as all are contemplated within the scope of the computing device of FIG. 10.

[0210] The interconnect system **1002** may represent one or more links or busses, such as an address bus, a data bus, a control bus, or a combination thereof. The interconnect system **1002** may include one or more bus or link types, such as an industry standard architecture (ISA) bus, an extended industry standard architecture (EISA) bus, a video electronics standards association (VESA) bus, a peripheral component interconnect (PCI) bus, a peripheral component interconnect express (PCIe) bus, and/or another type of bus or link. In some embodiments, there are direct connections between components. As an example, the CPU **1006** may be directly connected to the memory **1004**. Further, the CPU **1006** may be directly connected to the GPU **1008**. Where there is direct, or point-to-point connection between components, the interconnect system **1002** may include a PCIe link to carry out the connection. In these examples, a PCI bus need not be included in the computing device **1000**.

[0211] The memory **1004** may include any of a variety of computer-readable media. The computer-readable media may be any available media that may be accessed by the computing device **1000**. The computer-readable media may include both volatile and nonvolatile media, and removable and non-removable media. By way of example, and not limitation, the computer-readable media may comprise computer-storage media and communication media.

[0212] The computer-storage media may include both volatile and nonvolatile media and/or removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, and/or other data types. For example, the memory **1004** may store computer-readable instructions (e.g., that represent a program(s) and/or a program element(s), such as an operating system. Computer-storage media may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by computing device **1000**. As used herein, computer storage media does not comprise signals per se.

[0213] The computer storage media may embody computer-readable instructions, data structures, program modules, and/or other data types in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may refer to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, the computer storage media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.

[0214] The CPU(s) **1006** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **1000** to perform one or more of the methods and/or processes described herein. The CPU(s) **1006** may each include one or more cores (e.g., one, two, four, eight, twenty-eight, seventy-two, etc.) that are capable of handling a multitude of software threads simultaneously. The CPU(s) **1006** may include any type of processor, and may include different types of processors depending on the type of computing device **1000** implemented (e.g., processors with fewer cores for mobile devices and processors with more cores for servers). For example, depending on the type of computing device **1000**, the processor may be an Advanced RISC Machines (ARM) processor implemented using Reduced Instruction Set Computing (RISC) or an x86 processor implemented using Complex Instruction Set Computing (CISC). The computing device **1000** may include one or more CPUs **1006** in addition to one or more microprocessors or supplementary co-processors, such as math co-processors.

[0215] In addition to or alternatively from the CPU(s) **1006**, the GPU(s) **1008** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **1000** to perform one or more of the methods and/or processes described herein. One or more of the GPU(s) **1008** may be an integrated GPU (e.g., with one or more of the CPU(s) **1006** and/or one or more of the GPU(s) **1008** may be a discrete GPU. In embodiments, one or more of the GPU(s) **1008** may be a coprocessor of one or more of the CPU(s) **1006**. The GPU(s) **1008** may be used by the computing device **1000** to render graphics (e.g., 3D graphics) or perform general purpose computations. For example, the GPU(s) **1008** may be used for General-Purpose computing on GPUs (GPGPU). The GPU(s) **1008** may include hundreds or thousands of cores that are capable of handling hundreds or thousands of software threads simultaneously. The GPU(s) **1008** may generate pixel data for output images in response to rendering commands (e.g., rendering commands from the CPU(s) **1006** received via a host interface). The GPU(s) **1008** may include graphics memory, such as display memory, for storing pixel data or any other suitable data, such as GPGPU data. The display memory may be included as part of the memory **1004**. The GPU(s) **1008** may include two or more GPUs operating in parallel (e.g., via a link). The link may directly connect the GPUs (e.g., using NVLINK) or may connect the GPUs through a switch (e.g., using NVSwitch). When combined together, each GPU **1008** may generate pixel data or GPGPU data for different portions of an output or for different outputs (e.g., a first GPU for a first image and a second GPU for a second image). Each GPU may include its own memory, or may share memory with other GPUs.

[0216] In addition to or alternatively from the CPU(s) **1006** and/or the GPU(s) **1008**, the logic unit(s) **1020** may be configured to execute at least some of the computer-readable instructions to control one or more components of the computing device **1000** to perform one or more of the methods and/or processes described herein. In embodiments, the CPU(s) **1006**, the GPU(s) **1008**, and/or the logic unit(s) **1020** may discretely or jointly perform any combination of the methods, processes and/or portions thereof. One or more of the logic units **1020** may be part of and/or integrated in one or more of the CPU(s) **1006** and/or the GPU(s) **1008** and/or one or more of the logic units **1020** may be discrete components or otherwise external to the CPU(s) **1006** and/or the GPU(s) **1008**. In embodiments, one or more of the logic units **1020** may be a coprocessor of one or more of the CPU(s) **1006** and/or one or more of the GPU(s) **1008**.

[0217] Examples of the logic unit(s) **1020** include one or more processing cores and/or components thereof, such as Data Processing Units (DPUs), Tensor Cores (TCs), Tensor Processing Units (TPUs), Pixel Visual Cores (PVCs), Vision Processing Units (VPUs), Graphics Processing Clusters (GPCs), Texture Processing Clusters (TPCs), Streaming Multiprocessors (SMs), Tree Traversal Units (TTUs), Artificial Intelligence Accelerators (AIAs), Deep Learning Accelerators (DLAs), Arithmetic-Logic Units (ALUs), Application-Specific Integrated Circuits (ASICs), Floating Point Units (FPUs), input/output (I/O) elements, peripheral component interconnect (PCI) or peripheral component interconnect express (PCIe) elements, and/or the like.

[0218] In various embodiments, one or more CPU(s) **1006**, GPU(s) **1008**, and/or logic unit(s) **1020** are configured to execute one or more instances of regional path planner **122** and/or speed decision maker **124**. Sorted measurements **292** generated by regional path planner **122** may then be used by speed decision maker **124** and/or additional components to perform additional processing such as planning and control functions.

[0219] The communication interface **1010** may include one or more receivers, transmitters, and/or transceivers that enable the computing device **1000** to communicate with other computing devices via an electronic communication network, included wired and/or wireless communications. The communication interface **1010** may include components and functionality to enable communication over any of a number of different networks, such as wireless networks (e.g., Wi-Fi, Z-Wave, Bluetooth, Bluetooth LE, ZigBee, etc.), wired networks (e.g., communicating over Ethernet or InfiniBand), low-power wide-area networks (e.g., LoRaWAN, SigFox, etc.), and/or the Internet. In

one or more embodiments, logic unit(s) **1020** and/or communication interface **1010** may include one or more data processing units (DPUs) to transmit data received over a network and/or through interconnect system **1002** directly to (e.g., a memory of) one or more GPU(s) **1008**.

[0220] The I/O ports **1012** may enable the computing device **1000** to be logically coupled to other devices including the I/O components **1014**, the presentation component(s) **1018**, and/or other components, some of which may be built in to (e.g., integrated in) the computing device **1000**. Illustrative I/O components **1014** include a microphone, mouse, keyboard, joystick, game pad, game controller, satellite dish, scanner, printer, wireless device, etc. The I/O components **1014** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, and touch recognition (as described in more detail below) associated with a display of the computing device **1000**. The computing device **1000** may include depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, touchscreen technology, and combinations of these, for gesture detection and recognition. Additionally, the computing device **1000** may include accelerometers or gyroscopes (e.g., as part of an inertia measurement unit (IMU)) that enable detection of motion. In some examples, the output of the accelerometers or gyroscopes may be used by the computing device **1000** to render immersive augmented reality or virtual reality.

[0221] The power supply **1016** may include a hard-wired power supply, a battery power supply, or a combination thereof. The power supply **1016** may provide power to the computing device **1000** to enable the components of the computing device **1000** to operate.

[0222] The presentation component(s) **1018** may include a display (e.g., a monitor, a touch screen, a television screen, a heads-up-display (HUD), other display types, or a combination thereof), speakers, and/or other presentation components. The presentation component(s) **1018** may receive data from other components (e.g., the GPU(s) **1008**, the CPU(s) **1006**, DPUs, etc.), and output the data (e.g., as an image, video, sound, etc.).

Example Data Center

[0223] FIG. **11** illustrates an example data center **1100** that may be used in at least one embodiment of the present disclosure. The data center **1100** may include a data center infrastructure layer **1110**, a framework layer **1120**, a software layer **1130**, and/or an application layer **1140**.

[0224] As shown in FIG. **11**, the data center infrastructure layer **1110** may include a resource orchestrator **1112**, grouped computing resources **1114**, and node computing resources (“node C.R.s”) **1116(1)-1116(N)**, where “N” represents any whole, positive integer. In at least one embodiment, node C.R.s **1116(1)-1116(N)** may include, but are not limited to, any number of central processing units (CPUs) or other processors (including DPUs, accelerators, field programmable gate arrays (FPGAs), graphics processors or graphics processing units (GPUs), etc.), memory devices (e.g., dynamic read-only memory), storage devices (e.g., solid state or disk drives), network input/output (NW I/O) devices, network switches, virtual machines (VMs), power modules, and/or cooling modules, etc. In some embodiments, one or more node C.R.s from among node C.R.s **1116(1)-1116(N)** may correspond to a server having one or more of the above-mentioned computing resources. In addition, in some embodiments, the node C.R.s **1116(1)-1116(N)** may include one or more virtual components, such as vGPUs, vCPUs, and/or the like, and/or one or more of the node C.R.s **1116(1)-1116(N)** may correspond to a virtual machine (VM).

[0225] In at least one embodiment, grouped computing resources **1114** may include separate groupings of node C.R.s **1116** housed within one or more racks (not shown), or many racks housed in data centers at various geographical locations (also not shown). Separate groupings of node

C.R.s **1116** within grouped computing resources **1114** may include grouped compute, network, memory or storage resources that may be configured or allocated to support one or more workloads. In at least one embodiment, several node C.R.s **1116** including CPUs, GPUs, DPUs, and/or other processors may be grouped within one or more racks to provide compute resources to support one or more workloads. The one or more racks may also include any number of power modules, cooling modules, and/or network switches, in any combination.

[0226] The resource orchestrator **1112** may configure or otherwise control one or more node C.R.s **1116(1)-1116(N)** and/or grouped computing resources **1114**. In at least one embodiment, resource orchestrator **1112** may include a software design infrastructure (SDI) management entity for the data center **1100**. The resource orchestrator **1112** may include hardware, software, or some combination thereof.

[0227] In at least one embodiment, as shown in FIG. **11**, framework layer **1120** may include a job scheduler **1133**, a configuration manager **1134**, a resource manager **1136**, and/or a distributed file system **1138**. The framework layer **1120** may include a framework to support software **1132** of software layer **1130** and/or one or more application(s) **1142** of application layer **1140**. The software **1132** or application(s) **1142** may respectively include web-based service software or applications, such as those provided by Amazon Web Services, Google Cloud and Microsoft Azure. The framework layer **1120** may be, but is not limited to, a type of free and open-source software web application framework such as Apache Spark™ (hereinafter “Spark”) that may utilize distributed file system **1138** for large-scale data processing (e.g., “big data”). In at least one embodiment, job scheduler **1133** may include a Spark driver to facilitate scheduling of workloads supported by various layers of data center **1100**. The configuration manager **1134** may be capable of configuring different layers such as software layer **1130** and framework layer **1120** including Spark and distributed file system **1138** for supporting large-scale data processing. The resource manager **1136** may be capable of managing clustered or grouped computing resources mapped to or allocated for support of distributed file system **1138** and job scheduler **1133**. In at least one embodiment, clustered or grouped computing resources may include grouped computing resource **1114** at data center infrastructure layer **1110**. The resource manager **1136** may coordinate with resource orchestrator **1112** to manage these mapped or allocated computing resources.

[0228] In at least one embodiment, software **1132** included in software layer **1130** may include software used by at least portions of node C.R.s **1116(1)-1116(N)**, grouped computing resources **1114**, and/or distributed file system **1138** of framework layer **1120**. One or more types of software may include, but are not limited to, Internet web page search software, e-mail virus scan software, database software, and streaming video content software.

[0229] In at least one embodiment, application(s) **1142** included in application layer **1140** may include one or more types of applications used by at least portions of node C.R.s **1116(1)-1116(N)**, grouped computing resources **1114**, and/or distributed file system **1138** of framework layer **1120**. One or more types of applications may include, but are not limited to, any number of a genomics application, a cognitive compute, and a machine learning application, including training or inferencing software, machine learning framework software (e.g., PyTorch, TensorFlow, Caffe, etc.), and/or other machine learning applications used in conjunction with one or more embodiments.

[0230] In at least one embodiment, any of configuration manager **1134**, resource manager **1136**, and resource orchestrator **1112** may implement any number and type of self-modifying actions based on any amount and type of data acquired in any technically feasible fashion. Self-modifying actions may relieve a data center operator of data center **1100** from making possibly bad configuration decisions and possibly avoiding underutilized and/or poor performing portions of a data center.

[0231] The data center **1100** may include tools, services, software or other resources to train one or more machine learning models or predict or infer information using one or more machine learning

models according to one or more embodiments described herein. For example, a machine learning model(s) may be trained by calculating weight parameters according to a neural network architecture using software and/or computing resources described above with respect to the data center **1100**. In at least one embodiment, trained or deployed machine learning models corresponding to one or more neural networks may be used to infer or predict information using resources described above with respect to the data center **1100** by using weight parameters calculated through one or more training techniques, such as but not limited to those described herein.

[0232] In at least one embodiment, the data center **1100** may use CPUs, application-specific integrated circuits (ASICs), GPUs, FPGAs, and/or other hardware (or virtual compute resources corresponding thereto) to perform training and/or inferencing using above-described resources. Moreover, one or more software and/or hardware resources described above may be configured as a service to allow users to train or performing inferencing of information, such as image recognition, speech recognition, or other artificial intelligence services.

Example Network Environments

[0233] Network environments suitable for use in implementing embodiments of the disclosure may include one or more client devices, servers, network attached storage (NAS), other backend devices, and/or other device types. The client devices, servers, and/or other device types (e.g., each device) may be implemented on one or more instances of the computing device(s) **1000** of FIG. **10**—e.g., each device may include similar components, features, and/or functionality of the computing device(s) **1000**. In addition, where backend devices (e.g., servers, NAS, etc.) are implemented, the backend devices may be included as part of a data center **1100**, an example of which is described in more detail herein with respect to FIG. **11**.

[0234] Components of a network environment may communicate with each other via a network(s), which may be wired, wireless, or both. The network may include multiple networks, or a network of networks. By way of example, the network may include one or more Wide Area Networks (WANs), one or more Local Area Networks (LANs), one or more public networks such as the Internet and/or a public switched telephone network (PSTN), and/or one or more private networks. Where the network includes a wireless telecommunications network, components such as a base station, a communications tower, or even access points (as well as other components) may provide wireless connectivity.

[0235] Compatible network environments may include one or more peer-to-peer network environments—in which case a server may not be included in a network environment—and one or more client-server network environments—in which case one or more servers may be included in a network environment. In peer-to-peer network environments, functionality described herein with respect to a server(s) may be implemented on any number of client devices.

[0236] In at least one embodiment, a network environment may include one or more cloud-based network environments, a distributed computing environment, a combination thereof, etc. A cloud-based network environment may include a framework layer, a job scheduler, a resource manager, and a distributed file system implemented on one or more of servers, which may include one or more core network servers and/or edge servers. A framework layer may include a framework to support software of a software layer and/or one or more application(s) of an application layer. The software or application(s) may respectively include web-based service software or applications. In embodiments, one or more of the client devices may use the web-based service software or applications (e.g., by accessing the service software and/or applications via one or more application programming interfaces (APIs)). The framework layer may be, but is not limited to, a type of free and open-source software web application framework such as that may use a distributed file system for large-scale data processing (e.g., “big data”).

[0237] A cloud-based network environment may provide cloud computing and/or cloud storage that carries out any combination of computing and/or data storage functions described herein (or one or

more portions thereof). Any of these various functions may be distributed over multiple locations from central or core servers (e.g., of one or more data centers that may be distributed across a state, a region, a country, the globe, etc.). If a connection to a user (e.g., a client device) is relatively close to an edge server(s), a core server(s) may designate at least a portion of the functionality to the edge server(s). A cloud-based network environment may be private (e.g., limited to a single organization), may be public (e.g., available to many organizations), and/or a combination thereof (e.g., a hybrid cloud environment).

[0238] The client device(s) may include at least some of the components, features, and functionality of the example computing device(s) **1000** described herein with respect to FIG. **10**. By way of example and not limitation, a client device may be embodied as a Personal Computer (PC), a laptop computer, a mobile device, a smartphone, a tablet computer, a smart watch, a wearable computer, a Personal Digital Assistant (PDA), an MP3 player, a virtual reality headset, a Global Positioning System (GPS) or device, a video player, a video camera, a surveillance device or system, a vehicle, a boat, a flying vessel, a virtual machine, a drone, a robot, a handheld communications device, a hospital device, a gaming device or system, an entertainment system, a vehicle computer system, an embedded system controller, a remote control, an appliance, a consumer electronic device, a workstation, an edge device, any combination of these delineated devices, or any other suitable device.

[0239] In sum, the disclosed techniques use a path and speed planning system for robot navigation. The system includes a global route planner, a regional path planner, and a speed decision maker. The global route planner generates a global route plan between start and destination locations that includes a sequence of waypoints. The regional path planner generates a regional path from a current location to a destination based on a next waypoint of the global route plan by using a route finding algorithm. The regional path avoids obstacles and is in a region that includes at least a portion of the global route plan. The size of the region can be based on the sensing range of the robot, for example. The regional path planner discretizes each region along the global route plan by dividing the region into discrete units in which locations can be specified. The discretization can use Frenet coordinates to identify the locations of the discrete units. Frenet coordinates specify locations as longitudinal and lateral displacements along a particular route, such as the global route plan, for example.

[0240] The regional path is generated by a path solver, which uses a path searching algorithm to search for a regional path based on a target waypoint, one or more candidate next movements, and movement costs determined for the candidate next movements. The target waypoint can be the next waypoint on the global route plan, or a different identified waypoint if a path to the next waypoint on the global route plan is not feasible. If the path solver fails to find a regional path, e.g., because the possible paths are occupied, then a failure handler can expand the search space and cause the path solver to search the expanded search space. Alternatively, the failure handler can cause the global route plan to be adjusted, and the regional path planner can search for a regional path to a new target waypoint on the new global route plan.

[0241] The speed decision maker generates a set of speed directives that specify how the speed of a robot is to be changed to conform to speed constraints associated with a regional path as the robot moves along the regional path. The speed control directives are generated based on the regional path, which can be from the path planner, a semantic map indicating the locations of objects along the regional path, and/or a distance map from a perception system. The speed constraints can be static constraints, such as reduced speed zones, or dynamic constraints that vary over time as a result of motion of obstacles such as automatic doors or pedestrians. The speed decision maker provides the speed directives to a trajectory planner, which generates an optimized trajectory for the robot by jointly optimizing the regional path and speed directives. The speed decision maker generates a speed constraint for each object of the respective type along the regional path. Each constraint can apply to a location along the regional path during a particular range of time. For

example, for an automatic door, a stopping constraint is generated when a door along the regional path is closed, and a speed zone constraint is generated to slow the robot when the door is open. The speed solver finds a sequence of speed control decisions that satisfy the constraints. The constraints impose costs when not satisfied, and the speed solver generates speed directives that minimize the overall constraint costs. There can be multiple interdependent constraints along a path, e.g., for multiple movable obstacles. If there are many interdependent constraints, the speed solver can search the combinations of paths and times permitted by the constraints to solve the constraints jointly.

[0242] One technical advantage of the disclosed techniques relative to the prior art is the ability to efficiently navigate large environments because fine-grain path planning is performed on individual regions instead of the substantially larger area between the start and destination locations. Further, Frenet coordinate discretization is used in the regions to improve the efficiency of path searching by focusing the path search on a space near the path. Accordingly, the disclosed techniques are faster and less resource-intensive than prior approaches that use a single path planner to generate the route. Another technical advantage of the disclosed techniques is the ability to dynamically adjust speed using semantic information from map labels and real-time perception, such as predicted positions of dynamic obstacles. Adjusting the speed in this way results in smoother robot motion, for example. Accordingly, the disclosed techniques enable the robot to navigate more safely and efficiently through the environment than prior approaches that do not use semantic information to dynamically adjust speed. These technical advantages represent one or more technological improvements over prior approaches.

[0243] 1. In some embodiments, a method comprises identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

[0244] 2. The method of clause 1, wherein a speed constraint of the one or more speed constraints is generated based at least on an obstacle type of a respective obstacle of the one or more obstacles.

[0245] 3. The method of any of clauses 1-2, wherein the speed limit specified by the speed constraint is determined based at least on the obstacle type of the respective obstacle.

[0246] 4. The method of any of clauses 1-3, wherein the obstacle type includes one or more of an automatic door, stairs, or a pedestrian.

[0247] 5. The method of any of clauses 1-4, wherein the speed profile of the robot to satisfies the one or more speed constraints when the speed profile minimizes a sum of costs determined by the one or more speed constraints based on the speed profile.

[0248] 6. The method of any of clauses 1-5, wherein the one or more obstacles include a dynamic obstacle identified by a distance map received from a perception system, wherein at least a portion of the dynamic obstacle moves over time in accordance with a motion description, and the speed limit is based on a location of at least a portion of the movable obstacle at a particular time.

[0249] 7. The method of any of clauses 1-6, wherein each speed constraint generated based at least on the dynamic obstacle specifies a spatial range to which the speed constraint applies, and wherein the spatial range is determined based at least on motion of at least a portion of the dynamic obstacle over time as specified by the motion description.

[0250] 8. The method of any of clauses 1-7, wherein each speed change decision of the one or more speed change decisions is generated using a respective speed constraint of the one or more speed constraints and specifies a speed-related action to be performed to cause a speed of the robot to satisfy the respective speed constraint.

[0251] 9. The method of any of clauses 1-8, wherein the speed-related action comprises one or

more of a stop action or an accelerate action.

[0252] 10. The method of any of clauses 1-9, wherein the speed-related action applies to the mobile robot if the mobile robot is located within a distance range of a given obstacle on the path during a time range, wherein the given obstacle, the distance range, and the time range are specified by the respective speed constraint.

[0253] 11. The method of any of clauses 1-10, further comprising generating, based at least on the one or more speed change decisions, one or more speed directives, each speed directive of the one or more speed directives specifying a distance range on the path, a time range, and a speed limit that applies to the mobile robot in the distance range during the time range.

[0254] 12. The method of any of clauses 1-11, wherein the generating the one or more speed change decisions comprises generating one or more candidate speed profiles, each candidate speed profile satisfying one or more static speed constraints of the one or more speed constraints, and each static speed constraint being generated based at least on an obstacle that does not move during at least a determined time period; determining, based at least on the one or more candidate speed profiles, whether to generate a stop decision for a dynamic speed constraint of the one or more speed constraints, the dynamic speed constraint being generated based at least on a dynamic obstacle; and in response to determining that a stop decision is to be generated for the dynamic speed constraint, generating the stop decision based at least on the dynamic speed constraint, the one or more speed change decisions including the stop decision.

[0255] 13. The method of any of clauses 1-12, wherein the generating the one or more speed change decisions comprises generating a representation of a path time-space region having a spatial dimension based at least on a length of the path and a time dimension based at least on a time of travel along the path, the path time-space region including a time-space representation of each speed constraint; searching the path time-space region for a particular speed profile that does not intersect the time-space representations of the speed constraints; and generating the one or more speed change decisions based at least on the particular speed profile.

[0256] 14. The method of any of clauses 1-13, wherein the identifying the particular speed profile that satisfies the speed constraints comprises identifying one or more path time-space intervals of the representation of the path time-space region; and searching each path time-space interval of the one or more path time-space intervals for a portion of the particular speed profile, the portion of the particular speed profile satisfying one or more particular speed constraints of the one or more speed constraints, and the one or more particular speed constraints having location ranges and time ranges that are in the interval.

[0257] 15. The method of any of clauses 1-14, wherein each of the one or more obstacles is within a threshold distance of at least one point on the path of the mobile robot.

[0258] 16. The method of any of clauses 1-15, wherein the speed profile represents values of the speed of the mobile robot during a time range.

[0259] 17. In some embodiments, one or more processors comprise processing circuitry to perform operations comprising identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

[0260] 18. The one or more processors of clause 17, wherein the one or more processors is comprised in at least one of a control system for an autonomous or semi-autonomous machine. a perception system for an autonomous or semi-autonomous machine. a system for performing simulation operations. a system for performing digital twin operations. a system for performing light transport simulation. a system for performing collaborative content creation for 3D assets. a system for performing deep learning operations. a system implemented using an edge device. a

system for generating or presenting at least one of virtual reality content, augmented reality content, or mixed reality content. a system implemented using a robot. a system for performing conversational AI operations. a system for performing one or more generative AI operations. a system implementing one or more large language models (LLMs). a system for generating synthetic data. a system incorporating one or more virtual machines (VMs). a system implemented at least partially in a data center. or a system implemented at least partially using cloud computing resources.

[0261] 19. In some embodiments, a system comprises one or more processors to perform operations comprising identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

[0262] 20. The system of clause 19, wherein the system is comprised in at least one of a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing simulation operations; a system for performing digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing deep learning operations; a system implemented using an edge device; a system for generating or presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system implemented using a robot; a system for performing conversational AI operations; a system for performing one or more generative AI operations; a system implementing one or more large language models (LLMs); a system for generating synthetic data; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

[0263] The disclosure may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant or other handheld device. Generally, program modules including routines, programs, objects, components, data structures, etc., refer to code that perform particular tasks or implement particular abstract data types. The disclosure may be practiced in a variety of system configurations, including hand-held devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The disclosure may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.

[0264] As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.

[0265] The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or

between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Claims

- 1.** A method, comprising: identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.
- 2.** The method of claim 1, wherein a speed constraint of the one or more speed constraints is generated based at least on an obstacle type of a respective obstacle of the one or more obstacles.
- 3.** The method of claim 2, wherein the speed limit specified by the speed constraint is determined based at least on the obstacle type of the respective obstacle.
- 4.** The method of claim 2, wherein the obstacle type includes one or more of an automatic door, stairs, or a pedestrian.
- 5.** The method of claim 1, wherein the speed profile of the robot to satisfies the one or more speed constraints when the speed profile minimizes a sum of costs determined by the one or more speed constraints based on the speed profile.
- 6.** The method of claim 1, wherein the one or more obstacles include a dynamic obstacle identified by a distance map received from a perception system, wherein at least a portion of the dynamic obstacle moves over time in accordance with a motion description, and the speed limit is based on a location of at least a portion of the movable obstacle at a particular time.
- 7.** The method of claim 6, wherein each speed constraint generated based at least on the dynamic obstacle specifies a spatial range to which the speed constraint applies, and wherein the spatial range is determined based at least on motion of at least a portion of the dynamic obstacle over time as specified by the motion description.
- 8.** The method of claim 1, wherein each speed change decision of the one or more speed change decisions is generated using a respective speed constraint of the one or more speed constraints and specifies a speed-related action to be performed to cause a speed of the robot to satisfy the respective speed constraint.
- 9.** The method of claim 8, wherein the speed-related action comprises one or more of a stop action or an accelerate action.
- 10.** The method of claim 9, wherein the speed-related action applies to the mobile robot if the mobile robot is located within a distance range of a given obstacle on the path during a time range, wherein the given obstacle, the distance range, and the time range are specified by the respective speed constraint.
- 11.** The method of claim 1, further comprising: generating, based at least on the one or more speed change decisions, one or more speed directives, each speed directive of the one or more speed directives specifying a distance range on the path, a time range, and a speed limit that applies to the mobile robot in the distance range during the time range.
- 12.** The method of claim 1, wherein the generating the one or more speed change decisions comprises: generating one or more candidate speed profiles, each candidate speed profile satisfying one or more static speed constraints of the one or more speed constraints, and each static speed constraint being generated based at least on an obstacle that does not move during at least a determined time period; determining, based at least on the one or more candidate speed profiles, whether to generate a stop decision for a dynamic speed constraint of the one or more speed constraints, the dynamic speed constraint being generated based at least on a dynamic obstacle; and in response to determining that a stop decision is to be generated for the dynamic speed constraint,

generating the stop decision based at least on the dynamic speed constraint, the one or more speed change decisions including the stop decision.

13. The method of claim 1, wherein the generating the one or more speed change decisions comprises: generating a representation of a path time-space region having a spatial dimension based at least on a length of the path and a time dimension based at least on a time of travel along the path, the path time-space region including a time-space representation of each speed constraint; searching the path time-space region for a particular speed profile that does not intersect the time-space representations of the speed constraints; and generating the one or more speed change decisions based at least on the particular speed profile.

14. The method of claim 13, wherein the identifying the particular speed profile that satisfies the speed constraints comprises: identifying one or more path time-space intervals of the representation of the path time-space region; and searching each path time-space interval of the one or more path time-space intervals for a portion of the particular speed profile, the portion of the particular speed profile satisfying one or more particular speed constraints of the one or more speed constraints, and the one or more particular speed constraints having location ranges and time ranges that are in the interval.

15. The method of claim 1, wherein each of the one or more obstacles is within a threshold distance of at least one point on the path of the mobile robot.

16. The method of claim 1, wherein the speed profile represents values of the speed of the mobile robot during a time range.

17. One or more processors comprising: processing circuitry to perform operations comprising: identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

18. The one or more processors of claim 17, wherein the one or more processors is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing simulation operations; a system for performing digital twin operations; a system for performing light transport simulation; a system for performing collaborative content creation for 3D assets; a system for performing deep learning operations; a system implemented using an edge device; a system for generating or presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system implemented using a robot; a system for performing conversational AI operations; a system for performing one or more generative AI operations; a system implementing one or more large language models (LLMs); a system for generating synthetic data; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.

19. A system comprising: one or more processors to perform operations comprising: identifying, using one or more maps of a physical environment, one or more obstacles associated with one or more portions of a path of a mobile robot in the physical environment; generating, based at least on the one or more obstacles, one or more speed constraints, each speed constraint specifying a speed limit for a respective portion of the path; and generating one or more speed change decisions specifying actions to be performed by the mobile robot to cause a speed profile of the mobile robot to satisfy the one or more speed constraints.

20. The system of claim 19, wherein the system is comprised in at least one of: a control system for an autonomous or semi-autonomous machine; a perception system for an autonomous or semi-autonomous machine; a system for performing simulation operations; a system for performing digital twin operations; a system for performing light transport simulation; a system for performing

collaborative content creation for 3D assets; a system for performing deep learning operations; a system implemented using an edge device; a system for generating or presenting at least one of virtual reality content, augmented reality content, or mixed reality content; a system implemented using a robot; a system for performing conversational AI operations; a system for performing one or more generative AI operations; a system implementing one or more large language models (LLMs); a system for generating synthetic data; a system incorporating one or more virtual machines (VMs); a system implemented at least partially in a data center; or a system implemented at least partially using cloud computing resources.
